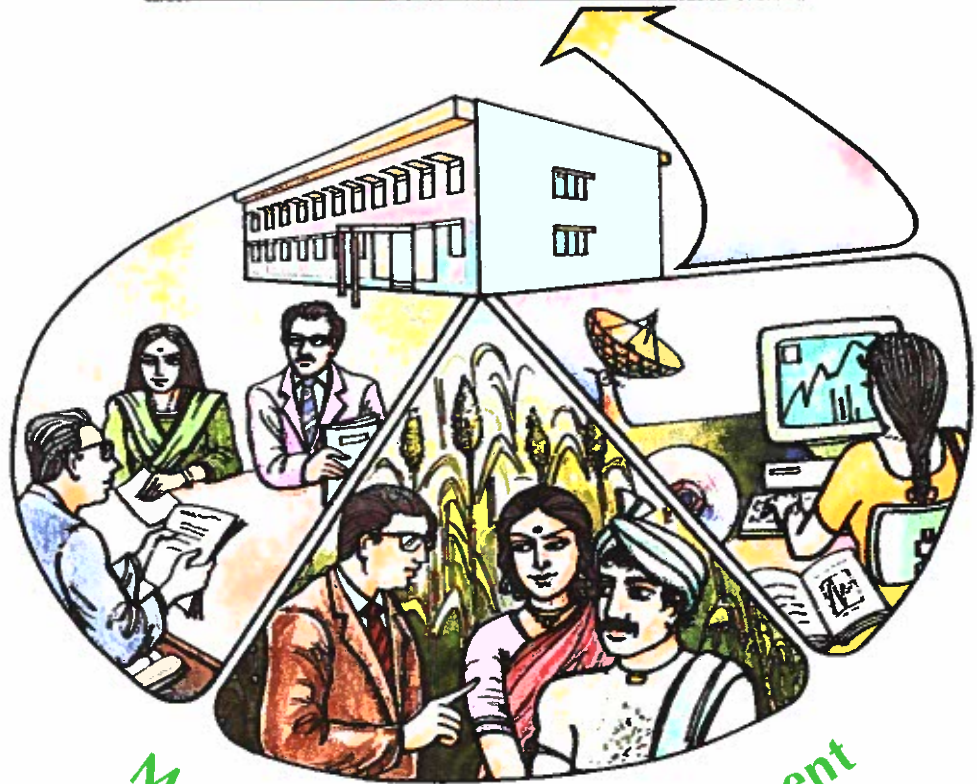


रा.कृ.अ.प्र.अ. वार्षिक प्रतिवेदन  
naarm annual report

1997-98



Management by empowerment

# वार्षिक प्रतिवेदन ANNUAL REPORT

1997-98



राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी  
राजेन्द्रनगर, हैदराबाद - 500 030, भारत

NATIONAL ACADEMY OF  
AGRICULTURAL RESEARCH MANAGEMENT  
Rajendranagar, Hyderabad - 500 030, INDIA

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Hyderabad.

## PREFACE

Since its inception, NAARM has taken up the onerous task of developing the vast human resource engaged in agricultural research, education and technology transfer functions in the country. To meet the emerging needs, NAARM initiated some new activities during 1997-98. As a first step toward recognition of its committed mandate, the Academy was restructured and the faculty members, on the basis of their specialization and sanctioned cadre strength, were grouped under three major divisions, viz., Agricultural Research Systems Management and Policies Division, Human Resource Development Division and Information and Communication Management Division. The Perspective Plan of the Academy was given a final shape by incorporating the suggestions made by experts and ICAR.

As a second step on introducing major reforms at NAARM, the content, substance and conduct of Foundation Course for the newly recruited Agricultural Research Service (ARS) Scientists of ICAR was revisited along with all the faculty members. Following a detailed introspection, it was accepted that ARS probationers on appointment to institutes have to work as team members with a moving spirit of 'provider of service'. In order to support this theme, FOCARS was redesigned to: (i) cover the transient phase from being a student to that of a would be scientist, (ii) emphasize skill development to become constructively involved in joint decision-making for managing agricultural research effectively and efficiently and, (iii) develop basic knowledge in service and financial rules. In addition to FOCARS, course curricula of senior level programmes were revamped appropriately so that each programme serves the client purpose best. Increase in the number of trainees and encouraging feedback are testimony of the positive response to the changes effected during the year under report.

A third initiative, which NAARM put its hands on during 1997-98, was to provide policy support to ICAR in the area of ongoing organizational and management reforms. In pursuance of this, a Workshop on Decentralization and Devolution of Powers in ICAR was organized for the senior executives and policymakers of the Council. Keeping in view the importance and topicality of the subject, this year's Annual Report carries the theme paper entitled "Decentralization and Devolution of Powers in ICAR" prepared by NAARM. Support was also provided to launch Project Based Budgeting in ICAR. Additional activities in the area of organizational and management reforms covered development of alternative annual assessment and research project file proformae.

In the process of revitalizing the Academy's programmes further, the fourth initiative centred around linkages with national and international centres pursuing mandates similar to that of NAARM. Reaffirmation of contacts with ISNAR - an international research centre promoting all aspects of human resource and institutional development - in the areas of

Agricultural Research and Information System Management and role of Distance Education in training of technology transfer personnel is worth mentioning. With firming up of the collaboration, it is our anticipated optimism that NAARM will gain high visibility in the region through improved programmes that serve the purpose and backstop the ongoing process of change in the NARS.

Organization of a series of faculty and executive development programmes for senior officers, professors and teachers of Tamil Nadu Agricultural University was another important milestone of 1997-98. For the specific benefit of academicians and teachers, NAARM with financial assistance from ICAR conducted a Summer School on Recent Advances in Pedagogy and Synergogy.

Deliberation on increasing research content of NAARM's training programmes was another important event of the year under report. In this endeavour, action through the mechanism of Research Advisory Committee was initiated to establish the research programmes in the broad area of agricultural research and education management and policy support to ICAR. Thirteen new projects were designed to refurbish and strengthen research base of NAARM.

Meteoric rise in the number of new initiatives would not have been possible but for the guidance and unstinted support given by Dr R.S.Paroda, Secretary, DARE and Director General, ICAR and Dr S.L.Mehta, Deputy Director General (Education). Professional support of Dr A.K.Sinha, Assistant Director General (Education) was highly valuable in timely conduct and launch of new programmes. While cooperation and help of NAARM faculty and other functionaries was exemplary, I specifically owe a gratitude to my colleagues Drs S.N.Saha, Jagannadham Challa, D.Rama Rao, T.Balaguru, P.Manikandan, R.Kalpana Sastry, A.Gopalam and Mr. B.Pandu Reddy who supported in preparation of this report. Assistance of M/s A.Vijayalakshmi, T.V.Ramadas, G.Aneeja, R.V.V.S.Prakasa Rao, P.Namdev and N.Narasimha Reddy in type setting, editing, printing is highly appreciated.

*J.C. Katyal*  
**J.C.Katyal**  
Director, NAARM

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J.C. Katyal  
J.C. Katyal  
Director, NAARM

गतिविधि

राज्य कृषि अनुसंधान प्रबंध अकादमी द्वारा अप्रैल 14 से 30, 1997 तक तमिलनाडु कृषि विश्वविद्यालय के अध्यापकों के लिए आयोजित "शिक्षण प्रौद्योगिकी में संकाय विकास कार्यक्रम" (प्राध्यापक अभिमुखीकरण शृंखला : बैच 1) का आयोजन किया गया। संकाय समस्या ने इसमें भाग लिया। डॉ. ए. गोपालम, डॉ. एस. एन. साहा और डॉ. जगन्नाथम चल्ता इस कार्यक्रम के समन्वयक थे।

गतिविधि

अप्रैल 3 से मई 3, 1997 तक "कृषि अनुसंधान परियोजना प्रबंध" (सी.ए.आर.पी.एम) पर दो सप्ताह के पाठ्यक्रम का आयोजन किया गया। इस कार्यक्रम में विभिन्न भा.कृ.अनु.प. संस्थानों और राज्य कृषि विश्वविद्यालयों (रा.कृ.वि) के 20 वरिष्ठ प्रबंधक और प्राध्यापकों ने भाग लिया। डॉ. के. वेंकटेश्वरु और डॉ. एस. एन. साहा इस कार्यक्रम के समन्वयक थे।

गतिविधि

मई 12 से 21, 1997 तक तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर के वरिष्ठ आचार्यों के लिए 10 दिवसीय "शिक्षण प्रौद्योगिकी में संकाय विकास कार्यक्रम" का आयोजन किया गया। इस कार्यक्रम में 20 वरिष्ठ आचार्यों ने भाग लिया। डॉ. एस. एन. साहा और श्री. एम. सुरेश कुमार इस पाठ्यक्रम के समन्वयक थे।

दो सप्ताहों में मई 19, 1997 को "पशुधन, पर्यावरण एवं मानव आवश्यकताओं का संतुलन" पर विचार विनियम हेतु पाठ्यक्रम का आयोजन किया गया। इस बैठक में भारत के विभिन्न क्षेत्रों से 32 प्रतिनिधियों ने भाग लिया। डॉ. अजीत पी. मारु इस कार्यक्रम के समन्वयक थे।

दो सप्ताहों में मई 19 - 21, 1997 तक फसल विज्ञान से संबंधित परियोजनाओं के प्रधान अनुसंधाताओं की एक विशिष्ट बैठक आयोजित की गई। इस बैठक में विभिन्न परियोजनाओं के 7 अनुसंधाताओं ने भाग लिया। इस बैठक के विचार-विनियम के सुविधक के रूप में डॉ. पी. एम. सुरेश कुमार, परियोजना परामर्शदाता, रोम में काम किया। डॉ. टी. बालगुरु ने इस बैठक के आयोजन में सहायता की।

गतिविधि

जून 9 - 30, 1997 तक "शिक्षा-शास्त्र और सहक्रिया-शास्त्र में अभिनव विकास" पर शोधकालीन पाठशाला का आयोजन किया गया। इस कार्यक्रम में रा.कृ.वि. और भा.कृ.अनु.प. संस्थानों के सत्रह सह-आचार्य और सहायक आचार्यों ने भाग लिया। इस शोधकालीन पाठशाला के डॉ. ए. गोपालम समन्वयक थे।

जून 11 - 21, 1997 तक "विंडोस एण्ड एम एस ऑफिस" पर एक प्राथमिक पाठ्यक्रम का आयोजन किया गया। इस पाठ्यक्रम में विभिन्न भा.कृ.अनु.प. संस्थानों के चौबीस वैज्ञानिकों ने भाग लिया। श्री. यू. मुरलीधर और श्री. एम. जे. प्रदीप कुमार इस पाठ्यक्रम के समन्वयक थे।

जून 21 से जुलाई 9, 1997 तक "कृषि में अनुसंधान एवं विस्तार कार्यक्रमों को विकसित करने में कृषि क्षेत्र संबंधी समस्याओं को पहचानना, विश्लेषण, विश्रान्त और सहभागी ग्रामीण मूल्यांकन तकनीक" पर दो सप्ताहीय प्रशिक्षण कार्यक्रम का आयोजन किया गया। इस कार्यक्रम में राज्य कृषि विश्वविद्यालय और भा.कृ.अनु.प. संस्थानों के सत्रह वरिष्ठ स्तर के संकाय सदस्यों और वैज्ञानिकों ने भाग लिया। डॉ. वी. ई. सबारत्नम और डॉ. पी. कुमार इस पाठ्यक्रम के समन्वयक थे।

जून 18-19, 1997 को कृषि शिक्षा पर डॉ. एम. एस. स्वामीनाथन समिति की तीसरी बैठक हुई। इस बैठक के अध्यक्ष डॉ. एम. एस. स्वामीनाथन और सदस्य-सचिव डॉ. एस. एल. मेहता, उपमहानिदेशक (शिक्षा), भा.कृ.अनु.प. थे।

### जुलाई

\* जुलाई 4-5, 1997 को सभी रा.कृ.वि. के बी.एससी (कृषि) के लिए स्नातकोत्तर पाठ्यचर्या की एकरूपता के बारे में विचार-विमर्श करने हेतु भा.कृ.अनु.प. शिक्षण प्रभाग द्वारा डॉ. एस. एल. मेहता, उप-महानिदेशक (शिक्षा) की अध्यक्षता में कृषि महाविद्यालयों के संकायाध्यक्षों हेतु दो दिवसीय बैठक बुलाई गई। इस कार्यक्रम में विभिन्न कृषि महाविद्यालयों के 35 संकायाध्यक्षों ने भाग लिया। डॉ. जगन्नाथम चल्ला इस कार्यक्रम के समन्वयक थे।

\* जुलाई 15-26, 1997 तक " कृषि अनुसंधान के लिए प्रबंध सूचना पद्धति " पर एक उन्नत पाठ्यक्रम चलाया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों, राज्य कृषि विश्वविद्यालयों और निगमों के 14 वैज्ञानिकों ने भाग लिया। डॉ. पी. मणिकण्डन और डॉ. टी. बालगुरु इस कार्यक्रम के समन्वयक थे।

\* जुलाई 22 से अगस्त 2, 1997 तक " कृषि में मानव संसाधन प्रबंध " पर दो सप्ताह के उन्नत पाठ्यक्रम का आयोजन किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों, रा.कृ.वि. और निजी एवं सार्वजनिक क्षेत्रों के अनुसंधान, विकास और शिक्षण संस्थानों से वैज्ञानिक, संकाय सदस्य और प्रशासकों और कृषि, पशुपालन, बागबानी और मात्स्यकी आदि, राज्य विभागों के अधिकारीगणों ने भाग लिया जिनकी संख्या 17 थी। इस पाठ्यक्रम के डॉ.एम. एम. अनवर और डॉ. कल्पना शास्त्री समन्वयक थे।

\* जुलाई 1-11, 1997 तक " डेटा विश्लेषण के लिए सांख्यिकीय साफ्टवेयर " पर एक पाठ्यक्रम आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. और रा.कृ.वि. के 13 वैज्ञानिकों ने भाग लिया। डॉ. के. विद्यासागर राव और श्री. के. वी. कुमार इस कार्यक्रम के समन्वयक थे।

\* जुलाई 28-30, 1997 तक " पुस्तकालय स्वचालन के लिए साफ्टवेयर पुनरीक्षा " पर तीन दिवसीय कार्यशाला का आयोजन किया गया। इस कार्यशाला में राष्ट्रीय कृषि अनुसंधान पद्धति के पुस्तकालयों एवं सूचना केंद्रों के प्रबंध से जुड़े हुए छियालीस वरिष्ठ पुस्तकालय अधिकारी और सूचना वैज्ञानिक, विक्रेता और प्रयोक्ताओं ने भाग लिया। श्री. वाई. वी. शिव प्रसाद और डॉ. अजीत पी. मारु इस कार्यशाला के समन्वयक थे।

### अगस्त

\* अकादमी में 11 से 27 अगस्त, 1997 तक तमिलनाडु कृषि विश्वविद्यालय कोयंबतूर के सहायक आचार्यों के लिए " शिक्षा प्रौद्योगिकी में संकाय विकास कार्यक्रम " पर दो सप्ताह का कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 34 सहायक आचार्यों ने भाग लिया। डॉ. जगन्नाथम चल्ला, श्री. बी. पाण्डु रेड्डी एवं डॉ. ए. गोपालम ने कार्यक्रम का समन्वयन किया।

\* 18-23 अगस्त, 1997 तक " कृषि अनुसंधान केंद्रों का प्रबंध " विषय पर एक सप्ताह का कार्यक्रम आयोजित किया गया। डॉ. वेंकटेश्वर्लु एवं डॉ. बी. एस. चन्देल ने कार्यक्रम का समन्वयन किया।

\* 22-23 अगस्त, 1997 तक " भारतीय कृषि अनुसंधान परिषद में अधिकारों का विकेंद्रीकरण " पर एक नीति परक पुनश्चर्या कार्यशाला आयोजित की गई। इस कार्यक्रम के विचार विमर्श की अध्यक्षता महानिदेशक डॉ. आर. एस. परोदा, जी ने की। इस कार्यशाला

हुई। इस बैठक के अध्यक्ष न.प. थे।

रूपता के बारे में विचार—  
क्षता में कृषि महाविद्यालयों  
याध्यक्षों ने भाग लिया। डॉ.

पाठ्यक्रम चलाया गया। इस  
[ डॉ. पी. मणिकण्डन और

ाठ्यक्रम का आयोजन किया  
विकास और शिक्षण संस्थानों  
य विभागों के अधिकारीगणों  
ने समन्वयक थे।

थ्यक्रम आयोजित किया गया।  
और श्री. के. वी. कुमार इस

वसीय कार्यशाला का आयोजन  
से जुड़े हुए छियालीस वरिष्ठ  
व प्रसाद और डॉ. अजीत पी.

त्यक आचार्यों के लिए "शिक्षा  
र्यक्रम में 34 सहायक आचार्यों  
मन्वयन किया।

का कार्यक्रम आयोजित किया

ण " पर एक नीति परक पुनश्चर्या  
परोदा, जी ने की। इस कार्यशाला

ने भा.कृ. अनु.प. (मुख्यालय) से सहायक महानिदेशक, भा.कृ.अनु.प. के विभिन्न संस्थानों के छः निदेशक तथा भा.कृ.अनु.प. के आठ  
सहायक निदेशक तथा वित्तीय सलाहकारों ने भाग लिया। डॉ. जे. सी. कत्याल निदेशक, डॉ. जगन्नाथम चल्ता, वरिष्ठ वैज्ञानिकों ने  
कार्यक्रम का समन्वयन किया।

#### दिसम्बर

विभिन्न संघतनों के सूचना केंद्र एवं पुस्तकालयों से जुड़े वरिष्ठ प्रबंधक अधिकारियों के लिए 2 से 12 सितंबर, 1997 तक  
"सूचना केंद्रों का प्रबंध" विषय पर दस दिवसीय पाठ्यक्रम का आयोजन किया गया। श्री. वाई. वी. शिवप्रसाद एवं डॉ. अजीत मारू  
कार्यक्रम का समन्वयन किया।

अनुसंधान संस्थानों में वीडियो तथा छायाचित्र के क्षेत्र में कार्यरत तकनीकी कर्मियों की कुशलता को विकसित करने हेतु 15—  
20 सितंबर, 1997 तक "कृषि क्षेत्र में वीडियोग्राफी" पर दो सप्ताह का पाठ्यक्रम आयोजित किया गया। श्री. के. आर. प्रभाकर  
जय श्री गोपाल जगन्नाथन राव ने इसका समन्वयन किया, जिसमें आठ सदस्यों ने भाग लिया।

25—26 सितंबर, 1997 को राष्ट्रीय कृषि प्रौद्योगिकी कार्यक्रमों के अन्तर्गत आयोजित अनुसंधान कार्यक्रमों के प्रधान  
अनुसंधानकर्ताओं के लिए एक "अभिमुखीकरण कार्यशाला" आयोजित की गई थी। राष्ट्रीय कृषि प्रौद्योगिकी कार्यक्रमों के अन्तर्गत  
आयोजित अनुसंधान कार्यक्रमों के प्रधान अनुसंधानकर्ताओं तथा कृषि राज्य विश्वविद्यालयों के 90 अनुसंधान निदेशकों ने इस कार्यक्रम  
में भाग लिया। श्रीमती संग्या शेनार्य तथा जी.एल. कौल, भा.कृ.अनु.प. ने इस कार्यशाला का समन्वयन किया।

कार्यक्रमों में 12 सितंबर, 1997 तक हिन्दी सप्ताह समारोह आयोजित किया गया।

अभ्युक्तों, कृषि विश्वविद्यालय के 22 संकाय सदस्यों के लिए 8 से 18 सितम्बर, 1997 तक शिक्षण प्रौद्योगिकी में संकाय  
विकास कार्यशाला आयोजित किया गया। डॉ. ए. गोपालम, डॉ. एस. एन. साहा एवं डॉ. जगन्नाथम चल्ता, इस कार्यक्रम के  
समन्वयक रहे।

#### नवम्बर

"विज्ञान प्रसारण प्रणाली ऑफिस" पर एक 10 दिवसीय प्रारंभिक पाठ्यक्रम 3 से 13 नवंबर, 1997 तक चलाया गया।  
कार्यशाला आयोजित करने में भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के वैज्ञानिक और कंप्यूटर से जुड़े कर्मिकों ने भाग लिया। डॉ. के.  
विजयलक्ष्मी और श्रीमती संग्या शेनार्य इस कार्यक्रम के समन्वयक थे।

"कृषि में जनसामान्यक मोटाला से जुड़े मुद्दे" पर एक तीन दिवसीय राष्ट्रीय कार्यशाला नवंबर 18—20, 1997 तक राष्ट्रीय  
कृषि विज्ञान संस्थान, पटना में आयोजित हुई। दिल्ली तथा विज्ञान और प्रौद्योगिकी विभाग, नई दिल्ली के सहयोग से आयोजित की गई। डॉ.  
के. विजयलक्ष्मी और श्रीमती संग्या शेनार्य इस कार्यशाला के समन्वयक थे।

"कृषि शिक्षण का प्रसार केंद्र कैसे हो?" पर दो दिवसीय कार्यशाला 24—25 नवंबर, 1997 को आयोजित की गई।  
कार्यशाला आयोजित करने में भा.कृ.अनु.प. के सहायक निदेशक और भा.कृ.अनु.प. शिक्षा विभाग के सदस्यों ने भाग लिया। इस कार्यशाला के समन्वयक  
डॉ. के. विजयलक्ष्मी और डॉ. ए. गोपालम थे।

\* राष्ट्रीय कृषि अनुसंधान पद्धति के स्कोट विश्लेषण पर एक उन्नत पाठ्यक्रम 25-28 नवंबर, 1997 तक आयोजित किया गया। जिसमें भा.कृ.अनु.प. के प्रधान वैज्ञानिक और वरिष्ठ वैज्ञानिक, अनुसंधान गत निदेशक और कृषि अनुसंधान केंद्रों के प्रभारी आचार्यों ने भाग लिया। इसमें सहभागियों की कुल संख्या 27 थी। इस कार्यक्रम के समन्वयक डॉ. के. वेंकटेश्वर्लु और डॉ. बी. एस. चन्देल थे।

\* " शिक्षा प्रशासन और प्रबंध का कार्यपालक विकास पाठ्यक्रम " 27 नवंबर से 2 दिसंबर, 1997 तक आयोजित किया गया। जिसमें तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर के 22 संकाय सदस्यों ने भाग लिया। इस कार्यक्रम के समन्वयक डॉ. जगन्नाथम चत्ता, डॉ. ए. गोपालम और डॉ. एस. एन. साहा थे।

### दिसंबर

\* " कृषि अनुसंधान की सूचना नीति " पर एशिया-एन ए आर एस के सूचना और जैव तकनीक के उत्तरदायित्व वाले वरिष्ठ नीति निर्माता और प्रबंधकों हेतु आई.सी.ए.आर - आई.एस.एन.ए.आर अन्तर्राष्ट्रीय कार्यशाला 8-11 दिसंबर, 1997 तक आयोजित की गई। इसमें चीन, मलेशिया, श्रीलंका और भारत के 27 सहभागियों ने भाग लिया। डॉ.टी. बालगुरु, इस कार्यशाला के समन्वयक थे।

\* " शिक्षा प्रशासन और प्रबंध " पर दो कार्यपालक विकास कार्यक्रम 18-23 दिसंबर, 1997 और 26-30 दिसंबर, 1997 तक आयोजित किए गए, जिसमें तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर के 37 वरिष्ठ कार्यपालकों ने भाग लिया। इसके समन्वयक डॉ. जगन्नाथम चत्ता, डॉ. ए. गोपालम और डॉ. एस. एन. साहा थे।

\* " राजभाषा कार्यान्वयन में निरीक्षण व्यवस्था " पर एक कार्यशाला दिसंबर 10-12, 1997 तक आयोजित की गई। इसमें कुल 22 सहभागियों ने भाग लिया, जिनमें सहायक निदेशक, तकनीकी अधिकारी तथा राजभाषा कार्यान्वयन से जुड़े अन्य कर्मचारी शामिल थे। इसके समन्वयक डॉ. ए. गोपालम और श्री. डी. वेंकटेश्वर्लु थे।

### जनवरी

पंजाब कृषि विश्वविद्यालय, लुधियाना के मानव संसाधन विकास केंद्र में 6-17 जनवरी, 1998 तक पंजाब कृषि विश्वविद्यालय लुधियाना के सहायक तथा सह आचार्य तथा महाराष्ट्र के चार राज्य विश्वविद्यालयों के कुल 6 सहभागियों के लिए कार्यक्रम आयोजित किया गया। भा.कृ.अनु.प. के कृषि मानव संसाधन विकास परियोजना के अन्तर्गत इस कार्यक्रम का समन्वयन डॉ. जगन्नाथम चत्ता एवं डॉ. ए. गोपालम ने संयुक्त रूप से किया।

### फरवरी

आचार्य एन.जी.रंगा कृषि विश्वविद्यालय, हैदराबाद के कुल 16 वरिष्ठ कार्यपालकों के लिए 2-11 फरवरी, 1998 तक " शिक्षा प्रशासन एवं प्रबंध पर कार्यपालक विकास कार्यक्रम " आयोजित किया गया। इस कार्यक्रम का समन्वयन डॉ. एस. एन. साहा एवं डॉ. ए. गोपालम ने किया।

### मार्च

आचार्य एन.जी.रंगा कृषि विश्वविद्यालय के 16 सहायक आचार्य तथा सहआचार्यों के लिए एस. वी. कृषि महाविद्यालय, तिरुपति में " शिक्षा प्रौद्योगिकी में संकाय विकास कार्यक्रम " आयोजित किया गया। डॉ. ए. गोपालम एवं डॉ. जगन्नाथम चत्ता इस कार्यक्रम के समन्वयक रहे।

## ACTIVITIES AT A GLANCE

### April

A two-week "Faculty Development Programme in Educational Technology" for the teachers of Tamil Nadu Agricultural University, Coimbatore, (Teacher Oriented Series : Batch 1) was organized from April 14 to 30, 1997 at NAARM. Twenty six faculty members attended the programme. Drs A. Gopalam, S.N.Saha, and Jagannadham Challa co-ordinated the programme.

A two-week course on "Agricultural Research Project Management (CARPM)" was organized from April 23 to May 3, 1997. Twenty senior-level managers and senior scientists from various ICAR institutes and State Agricultural Universities (SAUs) participated in the programme. Drs K. Venkateswarlu and S.N. Saha co-ordinated the programme.

### May

A ten-day "Faculty Development Programme in Educational Technology" was held for the senior professors of Tamil Nadu Agricultural University, Coimbatore from May 12 to 21, 1997. Twenty senior professors attended the programme. Drs S.N.Saha, A. Gopalam, and Mr. M. Suresh Kumar co-ordinated the programme.

A round table meeting on "Balancing Livestock, Environment and Human Needs" deliberation was put together at the Academy on May 19, 1997. Thirty two stakeholders and representatives drawn from different parts of the country, participated in the meeting. Dr Ajit K. Mann co-ordinated the meeting.

A specialized meeting of principal investigators of projects pertaining to crop sciences was arranged at the academy from May 19 to 21, 1997. About seven principal investigators of different projects participated in the meeting. The deliberations in the meeting were facilitated by Dr Peter Smith, Project Management Consultant, Rome. Dr T. Balaguru back supported the meeting.

### June

A Summer School on "Recent Advances on Pedagogy and Synergogy" was launched from June 9 to 30, 1997. Seventeen Associate Professors and Assistant Professors from various SAUs and ICAR Institutes participated in the programme. Dr A. Gopalam acted as the Director of the Summer School.

An introductory course on "Windows and MS Office" was arranged from June 11 to 24, 1997. Twenty four scientists from various ICAR institutes participated in the course. Mr H. Menadhar and Mr M.K. Pradeep Kumar co-ordinated the course.

\* A two-week training programme on "Rapid, Relaxed, and Participatory Rural Appraisal Techniques for Identifying Field Problems to Develop Research and Extension Programmes in Agriculture" was organized from June 24 to July 9, 1997. Seventeen senior-level faculty members and scientists of State Agricultural Universities and ICAR institutes participated in the programme. Drs VE. Sabarathnam and P. Kumar were the course co-ordinators.

\* The third meeting of Dr M.S. Swaminathan Committee on Agricultural Education was held on June 18 and 19, 1997. The meeting was chaired by Dr M.S. Swaminathan. Dr S.L. Mehta, DDG (Edn.), ICAR, acted as the Member-Secretary.

### July

\* A two-day meeting of the Deans of Agricultural Colleges was arranged on July 4 and 5, 1997, by the education division of ICAR under the chairmanship of Dr S.L. Mehta, DDG (Edn) to discuss the unification of UG curriculum for B.Sc. (Ag) in all SAUs. Thirty five Deans of various Agricultural Colleges, participated in the programme. Dr Jagannadham Challa co-ordinated the programme.

\* An advanced course on "Management Information System for Agricultural Research" was started from July 15 to 26, 1997. Fourteen scientists from ICAR institutes, State Agricultural Universities, and Corporate Sectors participated in the programme. Drs P. Manikandan and T. Balaguru co-ordinated the programme.

\* A two-week advanced course on "Management of Human Resources in Agriculture" was organized from July 22 to August 2, 1997. Seventeen scientists, faculty members, and administrators from ICAR institutes, SAUs and from other research, development, and educational institutes in the private and public sectors, and officers of state departments of agriculture, animal husbandry, horticulture, and fisheries participated in the programme. The course was co-ordinated by Drs M. M. Anwer and R. Kalpana Sastry.

\* A course on "Statistical Softwares for Data Analysis" was put together from July 1 to 11, 1997. Thirteen senior scientists of the ICAR and SAUs participated in the programme. Dr K. Vidyasagar Rao and Mr K.V. Kumar were the co-ordinators of the programme.

\* A three-day workshop on "Software Review for Library Automation" was conducted from July 28 to 30, 1997. Forty six senior library officers and information scientists, vendors and users involved in management of libraries and information centers in NARS participated in the workshop. Mr Y.V. Siva Prasad and Dr Ajit P. Maru co-ordinated the workshop.

### August

\* A two-week "Faculty Development Programme in Educational Technology" was set up for the Assistant Professors of Tamil Nadu Agricultural University, Coimbatore, at the

Participatory Rural  
Arch and Extension  
Seventeen senior-  
and ICAR institutes  
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Dr S.L. Mehta, DDG  
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Academy, from August 11 to 27, 1997. Thirty four Assistant Professors attended the programme. Dr Jagannadham Challa, Mr B. Pandu Reddy, and Dr A. Gopalam co-ordinated the programme.

A one week programme on "Management of Agricultural Research Stations" was organized from August 18 to 23, 1997. Drs K. Venkateswarlu and B.S. Chandel co-ordinated the programme.

A policy level retreat workshop on "Decentralization and devolution of powers in ICAR" was arranged on August 22 and 23, 1997. Dr R.S. Paroda, Director General, steered the deliberations of the programme. Two ADGs from ICAR headquarters, six Directors from different ICAR institutes, eight Deputy Directors General and Financial Advisor of ICAR participated in the workshop. Drs J.C. Katyal and Jagannadham Challa co-ordinated the workshop.

#### September

A ten day course on "Management of Information Centres" was put together from September 2 to 12, 1997, for nine senior officers managing information centres and libraries in various organizations. Mr Y.V. Siva Prasad and Dr Ajit Maru co-ordinated the programme.

A two week course on "Videography in Agriculture" was arranged from September 13 to 27, 1997, for developing skills in the technical personnel engaged in videography and photography in research institutes. Eight members attended the course co-ordinated by Mr E. H. Prabhakar and Mr Ch. Janardhan Rao.

An orientation workshop for the Principal Investigators of research programmes under NATP was held on September 25 and 26, 1997. About ninety Directors of Research at IARs and Principal Investigators of various research programmes under NATP participated in the workshop. Dr Sandhya Shenoy and Dr G.L. Kaul, ICAR, co-ordinated the workshop.

Four week celebrations were marshalled at the Academy from September 8 to 12, 1997.

A Faculty Development Programme in Educational Technology" was organized for twenty two faculty members of TNAU from Sept. 8 to 18, 1997. Drs A. Gopalam, S.N. Saha and Jagannadham Challa were the co-ordinators.

#### November

A ten day course entitled "Introductory Course on "Windows and MS Office" commenced from November 3 to 13, 1997. Twenty nine participants, including scientists and

personnel engaged in computer operations from ICAR institutes and SAUs participated in the course. Dr K. Vidyasagar Rao and Mr M.J. Pradeep Kumar co-ordinated the programme.

\* A three-day national workshop on "Issues of Professional Women in Agriculture" was held from November 18 to 20, 1997, in collaboration with National Commission for Women (NCW), New Delhi, and Department of Science and Technology, New Delhi. Drs D. Rama Rao and N. Sandhya Shenoy were the co-ordinators of the workshop.

\* A two-day workshop on "How to Manage Agricultural Education Better?", was arranged on November 24 and 25, 1997. About twenty seven Deans of various SAUs and the members from the Education Division of ICAR participated in the workshop. Drs S.N. Saha and A. Gopalam co-ordinated the workshop.

\* An advanced course entitled "SWOT Analysis of National Agricultural Research System" was organized from November 25 to 28, 1997, to principal scientists and senior scientists of ICAR, Associate Directors of Research, and Professor In-charge of Agricultural Research Stations numbering 27, participated in the workshop which was co-ordinated by Drs K. Venkateswarlu and B.S. Chandel.

\* An "Executive Development Programme on Educational Administration and Management" was started from November 27 to December 2, 1997, for twenty two faculty members of Tamil Nadu Agricultural University, Coimbatore. Drs Jagannadham Challa, A. Gopalam and S.N. Saha were the co-ordinators of the programme.

### December

\* ICAR-ISNAR International workshop on "Information Policy for Agricultural Research" was held from December 8-11, 1997, for Senior Policy Makers and Managers with responsibilities for information and biotech management in Asian NARS. Twenty seven delegates from China, Malaysia, Sri Lanka and India participated in the workshop. Dr T. Balaguru co-ordinated the programme.

\* Two Executive Development Programmes on "Educational Administration and Management" were setup from December 18 to 23, 1997 and from December 26-30, 1997, for the 37 senior executives of Tamil Nadu Agricultural University, Coimbatore. Drs Jagannadham Challa, A.Gopalam and S.N.Saha co-ordinated the programmes.

\* A workshop on "Inspection System in the Implementation of Official Language" was conducted from December 10 to 12, 1997. Twenty Assistant Directors, Technical Officers, and other officials, working in the area of implementation of official language in ICAR institutes, participated in the workshop. Dr A. Gopalam and Mr D. Venkateswarlu were the co-ordinators of the workshop.

Participants participated in the  
led the programme.

Women in Agriculture"  
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Technology, New Delhi.  
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Jagannadham Challa,  
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search and Managers with  
ICARS. Twenty seven  
workshop. Dr T. Bala-

Administration and  
Management 26-30, 1997,  
Chennai. Drs Jagam-  
manna.

Official Language" was  
of, Technical Officers,  
Official language in ICAR  
Kateswarlu were the

A training programme on "Technological Forecasting and Assessment" was  
instituted from December 8 to 16, 1997, in collaboration with TIFAC, New Delhi. Fifteen  
scientists of ICAR and SAUs participated in the programme. Drs D. Rama Rao and  
S. B. Thanda co-ordinated the programme.

#### January

A "Faculty Development Programme in Educational Technology" was set up for 26  
Associate Professors and Associate Professors of PAU, Ludhiana and four SAUs of Maharashtra  
from January 6 to 17, 98 at HRD centre of PAU, Ludhiana. Drs Jagannadham Challa and  
A. Gopalam co-ordinated the programme under the aegis of AHRD project of ICAR.

#### February

An "Executive Development Programme on Educational Administration and Man-  
agement" was organized from February 2 to 11, 1998 for sixteen senior executives of  
AHRD, Hyderabad. Drs S.N. Saha and A. Gopalam co-ordinated the programme.

#### March

A "Faculty Development Programme in Educational Technology" was put together  
for sixteen Associate Professors and Associate Professors of ANGRAU at SV Agricultural College  
at Tirupati campus. Drs A. Gopalam and Jagannadham Challa were the co-ordinators.

### Summary of 1997-98 programmes organized by NAARM

|                             | 1997-98  |     |     |     |     |     |     |     |     |          |     |     |
|-----------------------------|----------|-----|-----|-----|-----|-----|-----|-----|-----|----------|-----|-----|
|                             | APR 1997 | MAY | JUN | JUL | AUG | SEP | OCT | NOV | DEC | JAN 1998 | FEB | MAR |
| Summer Field Programmes     |          |     |     |     |     |     |     |     |     |          |     |     |
| Workshops                   |          |     |     |     |     |     |     |     |     |          |     |     |
| Meetings                    |          |     |     |     |     |     |     |     |     |          |     |     |
| Orientation of Participants |          |     |     |     |     |     |     |     |     |          |     |     |
| Field Visits                |          |     |     |     |     |     |     |     |     |          |     |     |

## INTRODUCTION

Science and technology are being increasingly harnessed for augmenting agricultural production in India. Agricultural scientists play a vital role in this noble endeavour. The formal university education makes the scientists specialized in their respective disciplines. Although subject specializations fills scientists with knowledge, it hardly prepares them to be good managers, who could plan and organize agricultural research, education and extension programmes more scientifically and efficiently. Majority of them, in fact, are yet to become conscious of the scientific management in agriculture. It has, thus, become important for all those concerned with agricultural research, education and development to be familiar with the principles of scientific management leading to new heights in programme effectiveness and performance efficiency.

The urgent need for an institutional set up to provide such an opportunity for the agricultural scientists was felt by the Government of India. The National Commission on Agriculture first recommended to establish an All India Institute of Agricultural Administration and Management for training agricultural scientists. The expert committee, appointed by the Government in 1973, endorsed the setting up of a National Staff College for Agriculture. Accepting this recommendation, the Indian Council of Agricultural Research (ICAR) established the Central Staff College for Agriculture (CSCA) at Hyderabad, in September 1976. Keeping in view the importance of its role at all India level and beyond, CSCA was rechristened as National Academy of Agricultural Research Management (NAARM) in 1979.

Since its inception, the Academy has taken up the responsibility of organizing training programmes, workshops and seminars in the areas related to agricultural research, technology transfer and education management for the scientists of the ICAR Institutes and State Agricultural Universities (SAUs). Investigations on management problems associated with agricultural research and education formed another major activity of the Academy. The results from such research effort formed the basis for training as well as for organizing workshops and seminars. The Academy has also been functioning as a repository of ideas and information, both at the national and international levels, and acting as a clearing house for dissemination and utilization of such information and knowledge.

The Academy is contributing significantly to generate a sense of fraternity and inculcates scientific culture among the agricultural scientists in the country. This unique institution has already established its credibility in the national and international arena. It is looked upon as a model by the developing countries and it is fast developing into a viable Regional Training Centre in this part of the world to train agricultural scientists and academicians in agricultural research and education management.

The Academy is located in the sylvan environs of Rajendranagar, in the premises of Acharya N.G. Ranga Agricultural University (ANGRAU), on the south-western periphery of Hyderabad city. The well-planned campus, spreading over 50 hectares, has several modern buildings including academic block, faculty-cum-laboratory building, administrative block, halls of residences, scientists' home, and several faculty and staff quarters.

## Mandate

To organize and conduct training programmes in agricultural research management for the scientists at various levels

To build up high quality resource material in agricultural research management based on actual field experience

To undertake systematic review and study of management problems of agricultural research institutes, programmes and systems

To plan, organize and conduct workshops and seminars in research management and educational technology

To organize, liaise and co-ordinate programmes of international co-operation in the field of agricultural research management.

To extend the State Agricultural Universities (SAUs) in developing Regional Centres of Management to cater to the needs of various Universities

Over the years, the Academy has created necessary expertise and infrastructure to fulfil the above mandate. As the years went by, the Academy has been able to formulate clear-cut objectives to fulfil the expectations of the National Agricultural Research System (NARS).

## Objectives

To organize foundation courses for the new entrants of scientists to the Agricultural Research Service of ICAR

To design and organize training programmes, workshops and seminars; and to develop high quality material in agricultural research and education management for the senior and middle level scientists of NARS

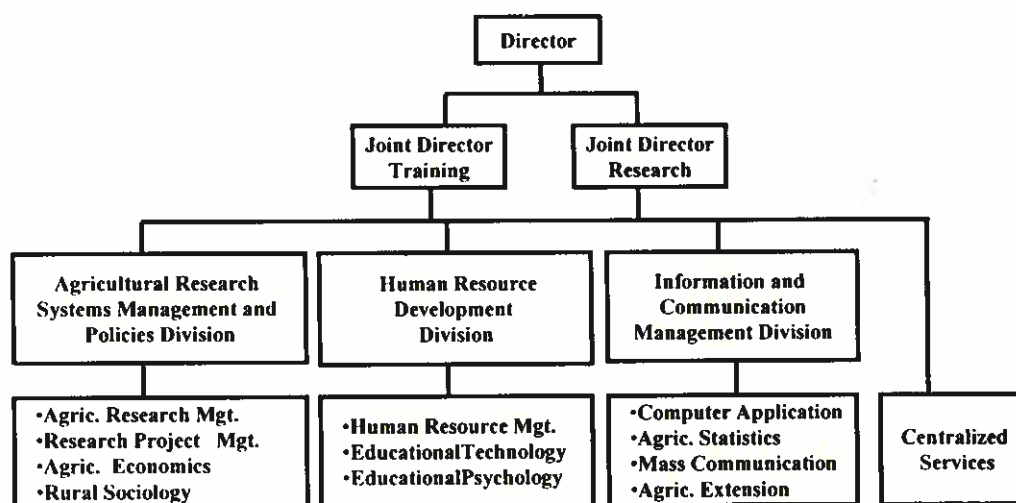
To play a major role in different processes of human resource development of NARS in the context of agricultural research and education

To undertake policy level studies on prioritization of research and resources allocation to enhance research productivity of the constituents of NARS and their programmes

To evolve suitable management techniques for effective planning, scheduling, coordination, monitoring, and evaluation of agricultural research and development projects in NARS

- ▲ To study the management issues of various programmes and patterns of technology transfer, develop effective system and strategies for effective communication of agricultural technologies
- ▲ To undertake development programmes for evolving more effective and efficient approaches, methods and models in higher agricultural education for use in SAUs and Deemed Universities of ICAR
- ▲ To promote and execute academic programmes of higher learning in the specialized areas of agricultural research and education management
- ▲ To develop expertise in information technology to meet the application software needs of NARS
- ▲ To undertake systematic review and develop appropriate systems for efficient and effective administrative and financial management in NARS
- ▲ To function as a repository of ideas and information in the form of database in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas
- ▲ To offer consultancy services in the field of agricultural research and education management, agricultural information and communication technology

## Organizational structure



The Academy is headed by a Director, who is guided and supported by the Institute Development Committee (IMC). The IMC reviews and promotes programmes of the Academy, and approves investments in new areas of research, education, training programmes, workshops, and seminars. In the area of research, the Research Advisory Committee (RAC) and the Research Council (SRC) provide broad guidelines and assist in developing and implementing specific programmes and projects. In the execution of various programmes, the Director is assisted by two Joint Directors (Currently one Joint Director is in position) and three Division Heads. The Academy is organized into three functional divisions viz., Agricultural Research Systems Management and Policies, Human Resources Development, and Information and Communication Management. A multi-disciplinary team of scientists, who also constitute the core faculty of the Academy, works in different divisions. Each faculty member is an agricultural scientist, with specialization in one of the disciplines of agriculture, who has also received advanced and specialized training in the broad area of management. As needed and appropriate, the Academy borrows services of outside specialist faculty for its multifarious training programmes. Functioning of the Academy is supported by a number of centralized services such as Administration and Accounts, Hostel Services, Health Services, and Press and Public Relations Cell. Current organizational structure of IIAARM is illustrated through an organogram.

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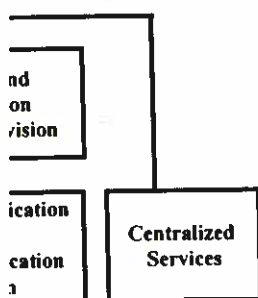
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### Decentralization and Devolution of Powers in ICAR\*

#### Preamble

With changing face of the national agricultural scenario in the country on the threshold of 21st century, and in the wake of globalization of agriculture, it becomes of paramount necessity to meet the challenges with confidence and competitiveness. The Indian economy is in an upswing, and it is inextricably linked to the growth in the agricultural sector. The growth in the agricultural sector would be driven by the compulsions for improving food and nutritional security and guided by the concern for environmental sustainability, poverty alleviation, employment generation, production by masses and profitability of farm enterprises. So that India remains below the Malthusian Cross (when the population food needs exceed the availability), the requirement of foodgrains is projected at 220 million tonnes for the year 2001, and at about 245 million tonnes for the year 2006. This is indeed a stiff challenge considering that the current production stands at about 195 million tonnes. The pressure is on ICAR since it is the premier organization vested with the primary responsibility of technology development. If past record is any indicator, there is no doubt on the ability of the ICAR to rise to the challenge.

Over the years, the ICAR has built a network of research and educational institutions, research stations, Krishi Vigyan Kendras (KVKs) and training centres. Specifically, during the last thirty years, considering the three elements of growth viz., a) expansion in terms of manpower and infrastructure, b) diversification from the point of national priorities and thrust areas, and c) modernization in view of latest methods of research and communication network, rise of ICAR as an organization of agricultural research, education and technology transfer has been far greater than in the past. The first two elements have been firmly anchored, and the Council is now in the process of modernization to attain global competitiveness.

During the current decade, a perceptible change in India's economy witnessed through foreign investments and enhanced domestic strength is visible. She is now experiencing the challenges of development, and it is imperative for the ICAR to realign its systems and processes for speedy development of technologies that match with the expectations of the government and people.

It is a historical fact that India was written off as a 'begging bowl' not so long ago. Indian agricultural scientists, labelled often of having low-esteem, however, rose to the challenge to the surprise of the developed nations and made India a leading producer of food. To sustain that spirit of achievement, there is an emergent need to put in rigorous efforts at 'renewal of

\* Presented by Dr J.C. Katyal, Director, NAARM in a Policy-level Workshop on "Decentralization and Devolution of Powers in ICAR", held at NAARM, Hyderabad on August 22-23, 1997.

## in ICAR\*

ICAR to overcome the seepage in of lethargy, complacency and inefficiency and to create a more responsive, viable and vibrant R & D organization.

I believe, there are five major challenges facing ICAR which must be addressed to urgently

1. To revitalize the system further to better growth of research output (through greater functional authority).

2. To enhance competitiveness of ICAR Institutes/Bureaus/Project Directorates/ National Research Centres/Zonal Research Stations/KVKs in terms of technology development and economic regeneration (through relevance, programme prioritization, efficiency and effectiveness).

3. To build synergy through convergence of institutes and multidisciplinary programmes (by building trans-institutional consortia supported by multidisciplinary teams of diverse discipline scientists pursuing time-bound mission programmes).

4. To introduce an element of transparency in performance and impact (through periodic performance monitoring and evaluation)

5. To create ICAR as a visionary organization which is future-driven (by building the capacity of resilience in that the entire system responds to the changing needs of Indian citizenry and her international obligations).

There is no reason not to take pride in the past achievements. However, it is now the time to recollect that pride by exalted scientific culture, stimulating environment and professional management systems with a perspective outlook. These challenges call for perspective building, restructuring, building decentralized and empowered systems of decision-making and operations, where scientists and other staff are encouraged for innovation, contribution, and are well motivated to build synergy through team-working and are fully accountable. To achieve this, there is no short cut except to change scientific mind-set and to bring a perspective shift in the systems of management and governance through dispersal of authority.

With this background in mind, the principal objectives for the deliberations are to share views and experiences, and generate ideas on authority delegation and decentralization through the measures on the following:

(i) decentralization and devolution of powers to enhance efficiency and responsiveness of ICAR setup.

(ii) To prevent development of any unethical power-centres by minimizing decentralized centralization along the administration hierarchy, and

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- iiii) to enhance coherence in the execution of delegated authority which matches with the responsibility in a system of accountability.

## Growth of ICAR

The Imperial Council of Agricultural Research was set-up on July 16, 1929, as a society. Following independence, it was renamed as the Indian Council of Agricultural Research (ICAR). Ever since, the functioning of ICAR has always been dynamic with periodic reviews conducted by several teams appointed from time to time. Apart from two Joint Indo-American teams of 1954 and 1959, Government of India set up an Agricultural Research Review Team in 1963 with a specific objective of enquiring into the agricultural research set-up and suggesting necessary changes for improvement. Based on these recommendations, ICAR assumed the administrative control of 9 research institutes during April, 1966 along with the Central Commodity Committees. Further in June 1970, certain proposals were approved by the Union Cabinet but before these could be implemented, Government of India appointed an Enquiry Committee headed by Hon'ble Shri P.G. Gajendragadkar in May 1972. In pursuance of the recommendations of the Gajendragadkar Committee, DARE was set up in December, 1973, and ICAR was retained as an autonomous society so as to confer necessary operational and functional self-governance in the organization of research and education. Management committees were set up at each institute and Agricultural Research Service was founded. During this period of reorganization, several functional and structural changes were made in the ICAR with adequate financial, administrative, operational and functional autonomy to discharge its responsibilities more effectively.

Subsequently in February 1981, another committee was set up under the Chairmanship of Late Shri Chandrasekhar Singh, Hon'ble Member of the then Parliament, to review the administrative and financial working of ICAR. Quite a few useful recommendations were made regarding delegation of powers to Directors and Management Committees at the institute level.

Another ICAR Review Committee was constituted under the Chairmanship of Dr G.V.K. Rao in April 1987 to review the then existing organizational infrastructure, personnel policies and functional role of ICAR. It is almost ten years since the report of GVK Rao Committee was submitted. Yet another expert report was submitted by the Tata Consultancy Services in 1995 with regard to the functioning of the ICAR Headquarters.

During the last ten years, ICAR grew phenomenally in terms of infrastructure expansion and inclusion of more and more areas of operation for research. The data in the following table indicate the trends in growth of number of research establishments during the last decade. It is evident that there were a total of 143 research units in 1987 in the form of National Institutes, Central Research Institutes, National Bureaux, National Research Centres, Project Directorates and AICRPs. Their number rose to 173 by 1997 indicating a decadal growth rate of approximately 21 per cent.

## ICAR Institutes\*

| Institutes                | 1987       | 1997       |
|---------------------------|------------|------------|
| National Institutes       | 4          | 5          |
| Central Institutes        | 35         | 40         |
| National Bureaux          | 4          | 4          |
| National Research Centres | 20         | 28         |
| Project Directorates      | 9          | 10         |
| AIICRP's                  | 69         | 84         |
| Others                    | 2          | 2          |
| <b>Total</b>              | <b>143</b> | <b>173</b> |

Units of Headquarters at Krishi Bhawan and Krishi Anusandhan Bhawan are not included.

The details of budgetary provisions are given in the following table to match the resources with the expansion.

## ICAR Budgetary Details 1987 - 1997 (Rs in crores)

|                 | 1986-87      | 1991-92       | 1996-97       |
|-----------------|--------------|---------------|---------------|
| <b>Non Plan</b> | <b>76.5</b>  | <b>158.33</b> | <b>226.00</b> |
|                 |              | 10.00         | 20.00         |
| <b>Plan</b>     | <b>81.0</b>  | <b>189.75</b> | <b>288.90</b> |
| <b>AP Costs</b> | <b>8.2</b>   | <b>14.00</b>  | <b>14.00</b>  |
| <b>Total</b>    | <b>165.7</b> | <b>372.08</b> | <b>548.90</b> |

These two tables of information, in no uncertain terms, point to the direction that unless a viable and professional system of decentralization and delegation of powers is evolved, it may be difficult to effectively and efficiently operate and manage the vast network of Institutes, Centres, Bureaux and Directorates with centralized control from the ICAR Headquarters at Krishi Bhawan, New Delhi. Since further expansion in the programmes is envisaged during the 9th Five Year Plan, it is all the more urgent that this issue is addressed in a more comprehensive and dispassionate way.

It will be naive to say that no efforts on decentralization and delegation of authority have been in place in the past. In fact, during the last couple of years, a perceptible change and process of renewal in the ICAR system is visible. Delegation of powers to the subject matter chiefs (DDGs) at the ICAR headquarters and more powers to the Directors of the Institutes, Project Co-ordinators and even Principal Investigators of the externally funded projects are among the bold decisions which ICAR has taken during the last three years. The background supporting this kind of initiatives was to:

- \* reduce the problems of communication and red tape,
- \* make quick and appropriate decisions,
- \* capitalize on the human resource available in the system,
- \* create a competitive climate within the organization,
- \* create a line of more capable managers, and
- \* diversify activities leading to convergence and synergy.

In spite of impending gains and enthusiastic initiatives and able leadership provided by Dr R.S.Paroda, the current Director General, for renewal of ICAR, there is still a lot of inertia in the system of governance and administration at various levels. The barriers to effective delegation of powers down the line creep up, perhaps, because of the following reasons:

- \* Fear of loss of power
- \* Notion that "I can do it better myself"
- \* Lack of confidence in subordinates
- \* Fear of being exposed
- \* Difficulty in briefing

Another important initiative taken by the present Director General is on the preparation of perspective plans "Vision 2020" for the next 25 years in respect of all the ICAR Institutes. This major exercise is indeed a laudable effort, as it is the first step in effective delegation and visionary planning. To realize the goals of Vision 2020 by each Institute and Project, it is imperative that the policy level functionaries of ICAR reflect on the past experiences and build further on system restructuring, decentralization and devolution of powers and responsibilities in a realistic mission-mode approach to prepare the ICAR system when it enters the next millennium. Ever since, Dr R.S.Paroda has taken over the leadership of ICAR, he has also launched a very important initiative of implementing several major recommendations made by GVK Rao Committee.

**Some of the observations and recommendations of GVK Rao Committee related directly to the subject of my presentation are reproduced here to share views and experiences on these.**

### **1. Autonomy and financial management of ICAR**

Para 5.16 and 5.17 of GVK Rao Committee Report

"The operational efficiency of any research organization is greatly enhanced by autonomous functioning with the minimum of bureaucratic interference. The reorganization

which the ICAR underwent in 1973 was, in fact, a major step taken to bestow such autonomy in its technical and administrative functioning. Though the ICAR was retained as a Scientific Society with the purpose of making it an autonomous body, today it does not exercise its autonomy because of the self-imposed existing practices of financial and procedural controls. Flexibility of operation, specially on financial matters, have become severely restricted as the ICAR follows the Government of India rules mutatis mutandis and has, more often than not, sacrificed its autonomy in favour of Government rules and regulations. It is interesting to note that the CSIR rules and regulations and byelaws also specify mutatis mutandis constraint, but its functioning is practically fully autonomous. We also found that several scientific organizations in the country like the CSIR institutes, BARC, Space Research Institute, Raman Research Institute, etc., functioned with a greater degree of autonomy, meeting the needs of scientific establishments".

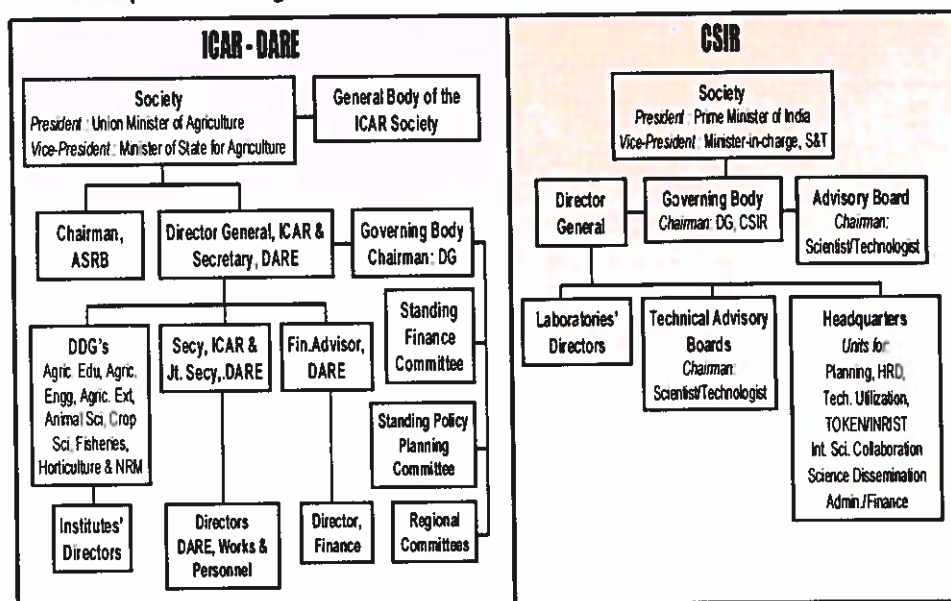
Unlike the Bhabha Atomic Research Centre, which was built up from a core group of scientists and administrators from the Tata Institute of Fundamental Research with its institutional and cultural practices, the ICAR was built up from the personnel from the Ministry of Food and Agriculture and is, thus, saddled with administrative practices prevalent in Government Departments. The cultural ethos of the ICAR is somewhat different from what is suited to an R & D organization. The operation mechanism of the ICAR includes factors such as control procedures, information system, reward-and-appraisal system, standardized rules and procedures and spatial arrangements".

It is clear from the above two paras and subsequent paras of GVK Rao Committee Report that ICAR continues to function as a government department in spite of autonomy in its technical and administrative functioning which may be due to "self-imposed existing practices". It is also very important to note that this committee has also extensively referred to the CSIR model of functioning. The structure of ICAR and CSIR are reproduced here for comparison and to discuss the merits and demerits of the existing structure and designations.

The ICAR, like the autonomy of ICAR. At present, Director General, ICAR; Secretary, ICAR and Financial Adviser (FA), ICAR are government officers of DARE holding three important positions of ICAR, whereas in CSIR the Director General and Joint Secretary (Admn) are holding additional charge of the Department of Industrial Research while FA, CSIR is a full time employee of CSIR. The structure of CSIR Headquarters is very slim with only 6 technical staff headed by scientists in the scale of Rs.5900-7300 while administration and finance are headed respectively by a Joint Secretary and a Financial Adviser. Director General is assisted by five Technical Advisory Boards chaired by eminent scientists/technologists who are not employees of CSIR.

GVK Rao Committee has suggested that the Director (F) should be upgraded to the rank of Joint Secretary and designated as Internal Financial Adviser and become a whole time employee of ICAR without any role for FA (DARE) in the internal affairs of ICAR [para 2.29 and (c)]

## Comparative organizational structure of ICAR-DARE &amp; CSIR



In the ICAR Headquarters, there are three important wings i.e. technical (subject matter) divisions headed by Deputy Directors General (DDGs), Administration and Finance being headed each, respectively by government officials. The functioning of the three wings needs to be deliberated upon to bring in more coherence through participatory functioning, transparency in decision-making and uninterrupted communication channels with minimal constraints and unreasonable delays.

Recently, through the efforts of present Director General the technical divisions headed by DDGs have been strengthened through decentralization of decision-making and delegation of powers as per recommendations of GVK Rao Committee. Ostensibly, these bold steps signal that integration has been brought between the technical and administration wings. In reality, there exists an unwritten knowledge and common experience that there are still a number of grey areas requiring indepth introspection. So that complete integration, based upon the principle of complementarity in place of competitiveness, is achieved both in letter and spirit and in accordance with the recommendations of GVK Rao Committee (para 5.37), institution of several new and forward looking decisions is necessary. The point for elaborate discussion and clarification is the functioning of Finance Wing, which remains parallel and independent. Furthermore, any decentralization and delegation processes for on-line senior functionaries shall be successful, only if the responsibilities are matched with the commensurate authority, failing which, it remains as a trap of decentralized-centralized system of operations, where rules and procedures without flexibility supercede the scientific effort and purpose.

## 2 Research management / administration

The management/administrative practices in ICAR are built around the principle of *mutatis mutandis* Government of India, which obviously leads to a strangle hold of bureaucracy over the R & D effort. The existing work-culture, environment and scientific practices are generally reflected by the prevailing systems of management and cultural practices, that are crafted by collective wisdom and experience either one way or the other. The philosophy of management by *mutatis mutandis* is either a bane or a blessing of the organization, needs introspection. If one likes to find excuses for failures, then *mutatis mutandis* is a blessing, and if one wants to resurrect the system as per the mandate of functional and social responsibilities, then it can prove to be the bane of the organizational performance. It is appropriate to recall the candid observations made by the GVK Rao Committee that "The cultural ethos of the ICAR is somewhat different from what is suited to an R & D organization" (para 5.17), whether this is close to or far from the truth needs deliberation and close scrutiny. These observations were made a decade ago yet their relevance and authenticity remains as intact today as ever. Apparently the system continued to remain placid for a long time.

At present, the Institutes are guided, serviced, reviewed and controlled by the Headquarters on all matters, which is not only time consuming but also increases the load of dependence on the Headquarters. There is absolute need to redefine the role of Headquarters in terms of promoting Institute based decision-making to facilitate more functional autonomy to the Institutes so that every little issue is not referred to Headquarters for deliberation. Since control of dependence of Institutes on Headquarters is so heavy that even trivial routine and un consequential issues are referred for approval/clearance. This has led to a state of affairs, wherein the Heads of the Institutes have a tendency to 'pass the buck' for decision making, or to delay the process of settlement of an issue which otherwise is within the realm of their authority. This particular phenomenon has assumed such serious proportions, that it has become the 'in-thing' of ICAR.

The very essence of decentralization needs to be redefined. The Headquarters alone cannot be blamed in this vexed issue, as the Institutes are also as much responsible for this malady. It is quite appropriate at this juncture that a fresh thinking reflects upon the past malady, and derives wisdom to fully derive benefits out of the existing decentralization of decision making and devolution of powers both at Headquarters and the Institutes. Any effort towards revitalizing the system through renewal process need not necessarily involve certain new structures, procedures and policy decisions. What is really needed is primarily a collective redefining of the rules, guidelines and executive orders issued by the Headquarters from time to time so that the scope for irresponsible individualism of the person occupying an important position can be minimized.

Nevertheless, there is also a need for certain structural and policy flexibility for strengthening the existing decentralization and delegation of powers so that certain minor

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irritants or constraints can be eliminated. In this regard, it may be appropriate to acquaint ourselves with the existing practices in CSIR.

### Bye-law 18 of CSIR

"CSIR Headquarters and Directors of National Laboratories would operate under the direct control of Director General without any intermediate line functionary. The laboratories would be freed from routine administrative and financial control of Headquarters. The Headquarters will, inter alia deal with the following matters:

a) Receipt and disbursement of funds from the government; b) Appointment of Director/Director level scientists; c) National missions; d) International and inter-agency scientific collaboration; e) Research fellowships and training; f) Liaison and interaction with major users and government departments; g) Data Bank of total CSIR R & D activities; h) Construction activity; i) Audit and vigilance matters; j) Grievance mechanism; k) Appellate against the decisions of Management Council and Directors; l) Parliament matters; m) Budget and Finance; n) Limitations imposed under the rules for delegation of financial powers; o) Accounting of funds to the Ministry of Finance/Accountant General and Public Accounts Committee; p) Recruitment / promotion policies and procedures; q) Common cadres; r) Legal matters; s) Patent matters; and t) Other matters as decided by DG".

One of the important recommendations made by the GVK Rao Committee was that the headquarters should decentralize the decision making process by giving more powers to the Institute Directors in specific areas (para 5.53). The committee also recommended that the Institute Management Committee (IMC) should have full responsibility with authority to deal with all operations and management matters, and be accountable for its actions and necessary technical, financial and executive powers should be vested with it (para 6.12). The above two recommendations are quite significant in terms of effective decentralization and delegation of responsibilities, powers and decision-making. Keeping in view the system in place, there is need to deliberate on these two issues at length and arrive at consensus for guidelines.

At this juncture, it is necessary to draw comparisons with the similar system in CSIR and hence, the powers and functions of Management Council are reproduced in Appendix I for reference. It is quite interesting to note that every laboratory in CSIR is governed by a Management Council and a Research Council. The Research Council is more or less an executive body having superior powers above the Management Council. In addition, the Research Council has power to approve and sanction budgetary proposals/items costing above Rs 50 lakhs, while Management Committee can approve proposals costing above Rs 5 lakhs and up to Rs 50 lakhs; while the Director can sanction up to Rs 5 lakhs. The most noticeable feature of these two bodies is that the proceedings are approved by the respective Chairperson. Another important virtue is that the Finance and Accounts Officer of the institute is a member of the Management Council.

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In the ICAR set-up, the institutes are governed by an Institute Management Committee (IMC) and Research Advisory Committee (RAC) which are, by and large, recommendatory in nature. Consequently, the proceedings are to be approved by the ICAR before recommendations are executed. The primary function of RAC is advisory in nature which is also limited to technical matters with no financial or administrative powers. The comparison of these two organizations (ICAR and CSIR) reveals that the existing systems within ICAR do not match in terms of decentralization of decision-making and delegation of powers. Another important feature is that the Finance and Accounts Officer (FAO) of the Institute is not a member of the IMC, whereas he is allowed to be a member for the IMC of another Institute. It is indeed a contentious issue that the FAO of the Institute has power to disagree with the decisions of the IMC and directly and confidentially communicate with the Director (F) recording his comments, which is nothing short of belying the authority and dignity of IMC. It is a classical example of a well intended check-point becoming a constraint strengthening the bureaucratic outlook of the existing system.

It is my considered opinion that there is an emergent need to carefully analyse past experiences and bring in the necessary modifications and delegation of authority to the RAC and IMC to develop autonomous functioning at Institute level with accountability.

It is also appropriate to mention here that CSIR is now embarking on a major policy decision of replacing Research Council set up of Laboratories with Executive Boards on the lines of corporate bodies and delegate greater powers and enforce accountability for each constituent Laboratory by 2001. At that point of time, each CSIR laboratory will be generating almost 50 per cent resource requirements through non-governmental funding.

### 1. System accountability

Individual and system accountability is the most important issue of organizational performance. Private sector attaches maximum value to monitoring and evaluation which serves as a basis for midcourse corrections of a project and for final performance assessment. In contrast, in the public sector, currently performance auditing, at the maximum, receives lip service. Even the well meaningful efforts to strengthen the standards of work in a sector like agricultural research have been far too meagre. In a vast network of ICAR employing several thousands of scientists, technicians, administration and accounts personnel in addition to the multitude of supporting staff, such a major systematic exercise to bring about system accountability has to be canvassed as a major priority effort. While policy, systems and infrastructure influence greatly the achievement of organizational goals, the general climate of competitiveness in the organization and the changes in the work attitude contribute significantly to optimization of research output by individuals and groups. A general system of scientific accountability which is both pragmatic and equitable can strengthen the climate of competitiveness and make the efforts for systematic decentralization and delegation of powers, very successful within ICAR. Overall scientific performance depends upon the collective system's thinking i.e., taking into consideration the sum total contribution of all the

support systems with a focus on the output of a project or the objective or the task. Accountability of research as a system cannot be measured alone or in isolation, while certain segments of the system remain immune to accountability. The attitude towards accountability requires a radical change through an inward look "at ourselves and our ways of functioning as a unit". Accountability ought to be the main theme of management in achieving goals of Perspective Plans in future. It is also important to visualise that accountability is nothing new but an effective system of leadership and management. It is rather an obligatory input to be inherently built in the system and if it is not, the entire research management endeavour itself is self-defeated. Hence, it is further emphasized for improving leadership and team-work capabilities, quality oriented systems and practices are needed in the ICAR system. Accountability determined by a non-subjective and non-illusory quantitative methodology is likely to bring about a perceptible change in the growth of scientific efficiency and effectiveness. As the funding crunch becomes tighter and tighter, emphasis on monitoring and evaluation of individual performance becomes greater and greater.

The issue of accountability is such a phenomenon that opinions differ, suggestions can be self-contradicting. Possibilities are that delicate situations may arise while dealing with certain sections of the organization. Also palpable resistance may be encountered and a lot of hypocrisy may set in, if it is not carefully planned, analysed and implemented. Hence, any method of accountability should specify the following conditions:

- \* Simplified so that it is easy to understand, implement, and is owned by all staff members.
- \* Rational, logical and orderly, so that personal bias and prejudice are minimised.
- \* Wholesome, such that all responsibilities are given due weightage.
- \* Systemic so that all sections and sub-systems are taken up.
- \* Transparent, responsive, and developmental in nature.

My entire purpose of emphasizing this particular issue on performance auditing is to make the existing system optimal and more responsive not only to the performance of a staff member but also addresses his developmental need also. Since the system I envisage stresses its ownership by staff members, what in fact I am proposing is centralized disempowerment. At this juncture, a renewed approach on accountability is necessary. Following five features may form the foundation of the policy guidelines of a performance management system.

1. Identification of goals/objectives/targets (a clear annual work plan is necessary).
2. Responsibilities are discussed, defined and delineated.
3. Performance measured in terms of quantifiable results against goals for the specific period.
4. Participation of all sections is obligatory.
5. Leadership support provided for counseling and individual development plan

The simplest import of this alternative performance management system is that leadership at various levels is unbiased, constructively critical and objectively crucial so that the overall responsibility to attain the agreed upon outputs lies with the management functionaries along the hierarchical toposequence. Apparently, the concept of decentralization and delegation of powers will require a shift to heterarchical system. A posteriori, it needs to be clearly understood that any system of decentralization or delegation of powers can only be implemented, if responsibilities assigned and expected outputs are matched with the commissioned authority. Hence, institution of a performance management/ accountability system is a must. For a sensible system of accountability assessment, it is essential to cultivate responsibility which is self-evident and open to auditing, supervision and control. Besides, it beholds and recognizes the need for a performance related incentive and reward system. Prima facie an accountability system must be as clear as glass. Above all, it has to be development oriented and not punitive. It will require a mind set change at all levels with Heads of institutes holding responsibility for overall performance and assessment. Finally, it is suggested that the issue of decentralization and delegation of powers in ICAR should be deliberated upon with concern for accountability built in the system for renewal of ICAR as conceived by the Director General.

### Other issues

It is also suggested that during the course of deliberations on decentralization and devolution of powers, there is need to look at certain contentious issues such as:

- Purchase procedures of stores/equipment
- Assessing/reviewing of unproductive activities/areas
- Project based budgeting
- Regular senior officers meetings at the institute level for reviewing and monitoring of ongoing programmes through participation and facilitation.
- Posting of administrative officers and finance and accounts officers in the same cadres at the institute level to bring more coherence.
- Empowerment to information on policy guidelines and executive orders of ICAR at the institute level and their uniform interpretation and application at various levels.

It is the desired objective of this gathering of policy level functionaries at this Academy to discuss thread-bare several issues of contention with a spirit of introspection through ways and means of functioning so as to resurrect and revitalise the organizational purpose without any ambiguity both at the Headquarters and at the Institute level. In the course of my presentation, I have raised a few issues referring to GVK Rao Committee recommendations, ICAR system of functioning and report of the Tata Consultancy Services so that a meaningful convergence of divergent functionaries for performance synergy can be brought in for reactivating the total ICAR setup. No doubt, it will be essential to shed lethargy and to prevent development of unethical power centres in the current system. Promotion of output-culture will necessitate to harness collective wisdom of and organization ownership by all concerned

for managing change through effective policy formulation, adoption and execution as ICAR marches ahead to meet the challenges beholden by its constituent Institutes in their 'Vision 2020' documents.

### Documents referred

1. CSIR Handbook, New Delhi, 1993.
2. ICAR Now and Ahead... June 1996.
3. Report of the ICAR Review Committee, 1988.
4. Memorandum of Association, Rules & Regulations and Bye-laws, CSIR, New Delhi, 1989.
5. Final Report on the ICAR Hqrs by Tata Consultancy Services, New Delhi, 1995.
6. National Agricultural Research, Education and Extension Education Systems in India. NAARM, 1988.

Appendix I

### Memorandum of Association

#### Rules & Regulations and Bye-laws of CSIR, New Delhi

For each National Laboratory, there would be a Research Council which would consist of the following:-

- a) Five external experts, one of whom shall be designated as Chairman.
- b) Representative of concerned scientific departments/agencies of Government of India.
- c) Director of the Laboratory.
- d) Senior Scientist from another National Laboratory.

Director General or his representative shall be a permanent invitee.

Members at (a) to (d) including the Chairman shall be nominated by Director General. The Secretary of the Research Council shall be nominated by the Director. The Secretariat for Research Council shall be provided by the concerned Laboratory.

The term of nominated members of Research Council shall be for a period of three years. The Research Council shall meet not less than two times in a year.

#### Functions of Research Council

Functions of Research Council shall be:

- i) Advise and recommend on the formulation of research programmes of the laboratory keeping the Five Year Plans and national priorities in view.

- (ii) Conduct periodic review of the research activities, assess the progress of the research programmes and advise on future directions.
- (iii) Advise on fostering linkages between the laboratory and academic institutions other research organizations, industry and potential clients.
- (iv) Any other function as may be assigned to it by the Director General/Governing Body.

The Director shall be guided by all decisions of the Research Council.

#### **Powers of Research Council**

Research Council shall have the power to:

- (i) constitute Selection Committees and Assessment Committees/Peer Groups for selection, merit and assessment promotions, from an approved panel of experts for all the S & T staff.
- (ii) recommend the resource allocation for major R&D activities of the Laboratory.
- (iii) approve of contract R & D Programmes beyond the powers of Directors.
- (iv) recommend devolution of necessary powers to the Project Leaders for the proper implementation of the research programmes.

The proceedings of the Research Council shall be approved by its Chairman. The Director General may review/amend any of the decisions of the Research Council which would be binding on it.

#### **Management Council (MC)**

There shall be a Management Council of each National Laboratory for management affairs of the Laboratory within the framework of Rules and Regulations, directions and guidelines issued by Director General/Governing Body and the Society. The Management Council shall consist of:

- (i) Director of the National Laboratory - Chairman
- (ii) Four Scientists of the Laboratory representing the staff of various age groups
- (iii) Two Director level Scientists from the same Laboratory/other laboratories
- (iv) Finance & Accounts Officer of the Laboratory
- (v) Controller of Administration/Administrative Officer of the Laboratory - Member Secretary

Director General or his nominee shall be a permanent invitee.

The Members at (ii) and (iii) will be nominated by the Director General. The term of the nominated members shall be for a period of three years. The Director General/ Governing Body have the powers to review/amend any of the decisions of the Management Council, which shall be binding on Management Council.

**Functions of the Management Council**

- a) To administer and manage the affairs and environs of the laboratory so as to support the Research Plan of the laboratory as approved by the Research Council.
- b) To act as an appellate authority for employees grievances.
- c) To write off irrecoverable losses of stores and money within the limits prescribed by CSIR.
- d) Such other functions and powers which are specifically not assigned to Research Council or Director.
- e) Any other matters as may be referred by Director/Director General, CSIR.

The proceedings of the Management Council shall be approved by its Chairman. The Governing Body or Director General as its Principal Executive Officer may review/amend any of the decisions of the Management Council, and pass such orders as considered necessary, which shall be binding on the Management Council".

## TRAINING PROGRAMMES

| Programme                                                                                       | Date                 | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Foundation courses</b>                                                                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Foundation Course for Agricultural Research Service                                             |                      | The foundation course for agricultural research service (FOCARS), a basic induction training for the newly recruited scientists to agricultural research service - lasts four months. It is organized in three phases viz., Phase 1 : Agricultural research systems and management processes - concept and principles; Phase 2 : Technology generation and transfer process - field experience training; and Phase 3 : Agricultural research systems and management processes - practice and applications. The foundation course aims at exposing the entry level scientists to various aspects of agricultural research project management, in addition to developing adequate skills in them for managing their research, by inculcating inter-disciplinary working in teams. It also prepares them to respond to the future demands of agricultural research management at the national and international level. |
| 01 FOCARS                                                                                       | 04-03-97<br>01-07-97 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 02 FOCARS                                                                                       | 03-06-97<br>30-09-97 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 03 FOCARS                                                                                       | 03-10-97<br>30-01-98 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 04 FOCARS                                                                                       | 07-01-98<br>05-05-98 | In all, 189 participants were trained through FOCARS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Foundation Course for Administrative Officers, Finance and Accounts Officers of ICAR Institutes | May 7-28, 1997       | This course was intended for the directly recruited Administrative Officers (AOs), and Finance and Accounts Officers (FAOs) of ICAR Institutes. The training contents included functioning of ICAR society, role of AOs and FAOs, fiscal and service rules, research project budgeting, etc. This course enabled the participants to understand the institutional research needs for a better interaction with scientists in changing situations of scientific needs and endeavour.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                 |                      | This course attracted 26 participants.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Senior-level programmes</b>                                                                  |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Course on Agricultural Research Project Management (CARPM)                                      | 23-04-97<br>03-05-97 | The main objective of this course was to expose the research project managers in the ICAR institutes, SAUs, and in other organizations to the concepts of agricultural research project and to the scientific management of agricultural research projects. The course dealt with topics like agricultural research project management, project planning, project selection, creativity and problem solving, systems approach in project planning, networks in project scheduling, etc.                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                                 |                      | Twenty participants were in attendance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Summer  
School on  
Recent  
Advances in  
Pedagogy  
and  
Synergogy

09-06-97  
29-06-97

The summer school exposed the participants to various advanced methods of instruction in Pedagogy, Andragogy and Synergogy for optimizing the learning- teaching effectiveness in the agricultural universities. Keeping in view the phenomenal growth of communication systems like satellite instructional T.V and fibro technology, multimedia, interactive video,expert systems, computer assisted instructional systems, etc. the instructional strategy focussed on the appropriateness and applicability with emphasis on suitable amendment and change for optimizing learning-teaching. This Summer School on Recent Advances in Pedagogy and Synergogy, thus, aimed to equip university teachers with various Pedagogical and Synergogical methods useful for optimizing learning-teaching in class room interactions.

Seventeen associate professors and assistant professors from State Agricultural Universities, and scientists from ICAR institutes participated in the programme.

Introductory  
Course on  
Windows and  
MS Office

11-06-97  
21-06-97  
03-11-97  
13-11-97

The introductory Course on Windows and MS Office was organized for exposing the scientists and computer technicians from ICAR and SAUs to windows operating environment and to the use of MS-Office software. Through intensive hands-on experience, the participants were familiarized with the use of Ms-Word for word processing, MS-Excel for spreadsheet, Powerpoint for graphics presentation, and MS-Access for database management.

Two such introductory courses were organized , the first one attended by 24, and the second one by 29 participants.

Rapid,  
Relaxed and  
Participatory  
Rural Appraisal  
Techniques for  
identifying field  
problems to  
develop  
research and  
extension  
programmes in  
Agriculture

24-06-97  
09-07-97

A programme on Rapid, Relaxed and Participatory Rural Appraisal Techniques for identifying field problems to develop research and extension programmes in Agriculture was organized. The specific objective of this programme was to make the participants competent in using the rapid and participatory rural appraisal techniques for the purpose of developing agricultural research and extension programmes.

The programme dealt with topics like Rapid, relaxed and participatory rural appraisal-concepts and differences; Principles of rapid and participatory rural appraisal techniques; Rapport building; Techniques of problem identification in field situation, Problem tree and intervention; Technology map; Social map,

Resource map; Matrix ranking and impact diagram; ITK map; Transact and Agroecology map; Climatic diagram; Mobility map; Farm household map; Wealth ranking formula and livelihood analysis in agriculture; Technology development; Technology assessment from farmers' point of view; Sustainability analysis.

Seventeen participants took part in this training.

Statistical 01-07-97  
Softwares for 11-07-97  
Data Analysis

The course on Statistical Softwares for Data Analysis was organized to provide an exposure to the basics of computer, to familiarize the scientists to work on statistical software packages for data analysis, and to provide adequate training on presentation of data and interpretation of results from the analysis. The course covered various areas like basic concepts of DOS operating system, hardware and software; statistical software packages of SPSS, MSTAT C, MICROSTAT, Lotus 1-2-3; Harvard graphics; and other related topics like GIS, remote sensing, GOST, etc.

The programme was attended by thirteen participants.

Management 15-07-97  
Information 26-07-97  
System for  
Agricultural  
Research

The course on Management Information System for Agricultural Research was conducted with a view to acquaint scientists with the concept of Management Information System (MIS) for agricultural research and assisting them to develop a computer-based MIS towards better research management in agriculture. As a part of the overall MIS, the course also exposed the participants to a project-based budgeting system (PBS). Through a project, the participants prepared the MIS for a small research institute and generated various reports for managerial decision making.

The course attracted 14 participants from ICAR Institutes and State Agricultural Universities.

Management 22-07-97  
of Human 02-08-97  
Resources in  
Agriculture

This course provided guidelines on the methodology to sensitise managers in agriculture to the issues of human behaviour in the organization and exposed the participants to the theories and practices of human behaviour in agricultural research environment for an in-depth understanding of individual, group and organizational behaviour.

About Seventeen Scientists, faculty members and administrators from ICAR Institutes, State agricultural Universities and other research, development and educational institutes in the private and public sector, officers of State Departments of Agriculture, Animal Husbandry, Horticulture and Fisheries from all over the country participated in the course.

Management  
of Agricultural  
Research  
Stations /  
Institutes

18-08-97  
23-08-97

The programme was organized to expose the participants to the concepts of management of agricultural research stations (MARS) and application of management approaches to these stations. The course covered various aspects like an overview of management of agricultural research stations, establishment of research stations and development, organization models of regional agricultural research stations (RARS), resource management, human transactions in MARS, research project planning and appraisal, research productivity and team work in experimental stations, etc. The programme attracted a whopping number of 34 participants.

Management  
of Information  
Centres

02-09-97  
12-09-97

This programme was organized to create awareness, knowledge enhancement and skills upgradation in managing information centres to officers managing information centres and libraries in various organizations a course on Management of Information Centres was organized. The course covers areas related to information management such as systems approach to information and information centres management, planning in information management, financial and administrative management of information centres, and organizational change and absorption of new information technology. Nine participants attended the course.

Videography  
in Agriculture

15-09-97  
27-09-97

The course was organized with the purpose of promoting communication technologies in the research institutions of ICAR and SAUs, exposing the participants to the technical aspects of videography, and introducing them to the emerging areas of videographic applications. Some of the topics covered included video recording system and format, principles of cinematography and photography, working of video cameras and VCR, studio and field recording, lighting in studio, etc.

The programme was attended by eight technicians from ICAR institutes.

Advanced  
course on  
SWOT  
analysis of  
NARS

25-11-97  
28-11-97

SWOT is the acronym for the process involving documentation and analysis of strengths, weaknesses, opportunities and threats and has recently found favours with researchers grappling with development issues of the organizations of agricultural sector. Application of SWOT analysis for agricultural research organization now assumes greater relevance because of internationalisation and globalization of agriculture.

The course attracted 27 participants from ICAR institutes, SAUs and non-governmental organizations.

Programme  
on technologi-  
cal forecasting  
and assess-  
ment

08-12-97  
16-12-97

The programme on technological forecasting and assessment was designed primarily to train the participants to use forecasting methods in education and research. The programme involved an inter-disciplinary format and its focus was on use and application of forecasting methods. The participants were trained on the principles, methods and application of forecasting techniques. Besides, the fifteen participants were provided literature and skills needed to offer a course on future studies and also computer programmes for qualitative and quantitative forecasting applications.

## Sponsored programmes

Faculty  
Development  
Programmes  
in Educational  
Technology

14-04-97  
30-04-97

12-05-97  
21-05-97

11-08-97  
28-08-97

08-09-97  
18-09-97

06-01-98  
17-01-98

09-03-98  
21-03-98

Considering the expertise developed at NAARM in the area of educational technology and management, it undertook a major HRD initiative on faculty development for SAUs, and the four deemed-to-be Universities of ICAR. During the period under report, six Faculty Development Programmes in Educational Technology for Tamil Nadu Agricultural University faculty members (Teacher-oriented series) were organized at the Academy. These programmes centred around four important entities of pedagogy, i.e., learning, teaching, testing and evaluation. All the academic sessions were conducted with theoretical overview, work-to-do sessions, group assignments, presentations and theme discussions.

In all, 144 faculty members were trained in the area of educational technology.

|                                                                               |          |
|-------------------------------------------------------------------------------|----------|
| Executive Development Programmes on Educational Administration and Management | 27-11-97 |
|                                                                               | 02-12-97 |
|                                                                               | 18-12-97 |
|                                                                               | 23-12-97 |
|                                                                               | 26-12-97 |
|                                                                               | 30-12-97 |
|                                                                               | 02-02-98 |
|                                                                               | 12-02-98 |

Four Executive Development Programmes on Educational Administration and Management were organized for the senior executives of the Tamil Nadu Agricultural University, Coimbatore. The main objective of these programmes was to develop management capacity in the area of agricultural administration and management. These programmes centred around four important entities of management, i.e., management of agricultural education, administration and finance, research management, and HRD strategies for education management. All the academic sessions were conducted with a theoretical overview, work-to-do sessions, group work, presentations, and theme discussions. The programme was organized in collaboration with the Tamil Nadu Agricultural University, Coimbatore, under the AHRD component. A total of 67 senior executive functionaries from TNAU participated in these programmes.

## Workshops

|                                        |          |
|----------------------------------------|----------|
| Software Review for Library Automation | 28-07-97 |
|                                        | 30-07-97 |

A three-day workshop on Software Review for Library Automation was organized to generate guidelines for selection of appropriate library software in India, and to provide specifications to NARS institutions for selecting appropriate software.

Forty six senior library officers and information scientists involved in management of libraries and information centres in NARS and related organizations participated in the workshop.

|                                                                                    |          |
|------------------------------------------------------------------------------------|----------|
| Policy Level Retreat Workshop on Decentralization and Devolution of Powers in ICAR | 22-08-97 |
|                                                                                    | 23-08-97 |

The Academy organised a Policy Level Retreat Workshop on "Decentralization and Devolution of Powers in ICAR". Directors from National Institutes of ICAR, seven Deputy Directors General, Financial Adviser, Director, DARE, Directors of some selected ICAR institutes, and two ADGs of ICAR participated in the workshop. Dr R.S. Paroda, Director General, ICAR, steered the deliberations of the workshop.

The major objective set for the workshop was to share the view points and experiences of the top-level functionaries of ICAR through interactive discussions on the following lines:

- \* Decentralization and delegation of powers to enhance efficiency and responsiveness of the ICAR set up.

## NAARM. HYDERABAD.



- \* To prevent development of any institutional power centres by administrative decentralization along the administrative hierarchy
- \* To enhance coherence in the exercise of delegated authority which must be with the responsibility.

The workshop provided a forum to the top management of ICAR to evolve policy directives on improving efficiency and effectiveness of the organizational renewal process by further strengthening co-ordination among various subject-matter divisions with administrative and financial wings through uninterrupted communication, participatory functioning and transparency in decision-making. The following issues were placed on the agenda for discussion in the workshop.

- \* Current delegation of powers to SMDs, its effectiveness in efficient research management
- \* Areas of convergence of administrative and financial authority to SMDs (convergence of trichotomy)
- \* Review of existing delegated authority dispensation
- \* Delegation of powers to institutes and expected responsibilities
- \* Decentralization and devolution of powers - cases and experiences
- \* *Mutatis-mutandis* - bane or blessing?
- \* Performance auditing of the assigned task/responsibility at various levels
- \* Project based budgeting, role of research advisory committee (RAC)
- \* Decentralization and delegation of powers -lessons from other R&D organizations

- \* Further empowerment of institute management committees (IMCs)
- \* Decentralization in ICAR - how far and how much?

Dr J.C. Katyal, Director, NAARM presented theme paper during the workshop. Drs G.B. Singh, O.S. Tomer, M.L. Madan, G.S. Shekawat and K. Gopakumar also presented statements. This policy level workshop was one of the series of retreat mode workshops being organised by NAARM in pursuance of the recommendations contained in the report by Dr A.L. Chowdary Committee (1995) on "Restructuring of training programmes of NAARM".

Orientation  
Workshop for  
the Principal  
Investigators  
of Research  
Programmes  
under NATP

25-09-97  
26-09-97

An Orientation Workshop for the Principal Investigators of Research Programmes under NATP was organized to sensitize the key functionaries/scientific leaders identified for respective programmes, about the implementation strategies to be adopted uniformly, and on the preparatory work to be completed by each leader before actual programme implementation.

Eighty Directors of Research of SAUs and Principal Investigators of various research programmes under NATP participated in the workshop.

Issues of  
Professional  
Women in  
Agriculture

18-11-97  
20-11-97

A three-day National Workshop on issues of Professional Women in Agriculture was organized at the Academy from November 18 to 20, 1997. The workshop aimed at identifying the issues and constraints in effective and efficient utilization of trained women graduates to meet the challenging social demand for agricultural development. Based on these objectives, six technical sessions were organized. Mrs Padma Seth, Adviser, National Commission for Women, New Delhi, inaugurated the workshop. Dr H.M. Eisa, Sr. Agriculturist, The World Bank, New Delhi; Dr Alexander Von der Osten, Executive Secretary, CGIAR, USA and Dr R.S. Paroda, Director General, ICAR attended the sessions on November 18, 1997 as resource persons.



The workshop was organized in collaboration with National Commission for Women, New Delhi; and Department of Science and Technology, New Delhi.

The recommendations emerged out of deliberations of the workshop stressed on education for women from rural areas, transfer of technology, managerial issues, entrepreneurship, personal policies, infrastructure and farm women issues.

Forty six Senior-level functionaries from developmental, education, research, financial and non-governmental sectors and organizations associated with agriculture and rural development participated in the workshop.

How to  
Manage  
Agricultural  
Education  
Better?

24-11-97  
25-11-97

A Workshop on "How to Manage Agricultural Education Better" was organized for deans of SAUs. Twenty seven Deans of various State Agricultural Universities and other members from the Education Division of ICAR participated in the workshop.



ICAR - ISNAR  
International  
Workshop on  
Information  
Policy for  
Agricultural  
Research

08-12-97  
11-12-97

A four-day International Workshop on Information Policy for Agricultural Research was organized at National Academy of Agricultural Research Management (NAARM), for senior Asian policymakers and managers with responsibilities for information and biotech management.

The main focus of the workshop was on Information Management / Information Technology for agricultural research. The emphasis throughout the programme was practical rather than academic. The workshop was organized in collaboration with International Service for National Agricultural Research (ISNAR), Japan International Research Center for Agricultural Sciences (JIRCAS) and Indian Council of Agricultural Research (ICAR), New Delhi. Twenty seven delegates from China, Malaysia, Sri Lanka and India participated in the workshop.

Inspection system in the implementation of official language 10-12-97  
12-12-97

A three-day workshop on Inspection systems in the implementation of official language was organized. The workshop dealt with various issues related to inspections by various committees for effective implementation of official language policies for the Hindi Officers, Assistant Directors (OL) and scientists in-charge of official language of various ICAR institutes. The workshop attracted 20 Hindi officers from various ICAR Institutes.

## Meetings

NATP Meeting 19-05-97  
21-05-97

Under the auspices of NATP, a specialized meeting of Principal Investigators of projects pertaining to crop sciences was organized at the Academy. Seven participants from UAS, Bangalore, CRIDA, Hyderabad and Project Director on Cropping System Research, Modipuram attended the meeting. The deliberations in the meeting were facilitated by Dr Peter Smith, Project Management Consultant, FAO, Rome.

Round Table Meeting on Balancing Livestock, Environment and Human Needs Interaction 19-05-97

A Round Table Meeting on balancing livestock, environment and human needs interaction was organized. The activity organized as a part of the global consultation for the electronic conference on balancing livestock, environment and human needs interaction. The main objectives of the meet were to identify strategies (policy, research, and extension) to alleviate the negative and enhance the positive impacts of livestock on natural resources in different types of livestock production systems in different countries and eco regions and also to identify areas of common interests which can lead to future research and development of institutional collaboration, partnership, net works, etc. 32 stakeholders and representatives drawn from different parts of the country, participated.

Deans of agricultural colleges meetings 04-07-97  
05-07-97

Thirty five Deans, Associate Deans, Principals of different agricultural colleges met, and discussed on unification of course curriculum of B.Sc. (Ag).

general, and, NAARM in particular, was explored further during the visit of Director General ISNAR to NAARM. It is foreseen that ICAR with a strong and diverse NARS and ISNAR with its wide experience in matters related to agricultural research management at the international level stand to gain through this cooperation.

Pending formal arrangements with ISNAR on the creation of this centre, some specific areas have been identified for cooperation. These include:

- Distance education in agricultural research management
- Management of information system
- Performance assessment and accountability enhancement of agricultural research
- Research station management
- Training for HRD

## Felicitation to Padmabhushan Dr R.S. Paroda

Dr R. S. Paroda, Secretary, Department of Agricultural Research and Education (DARE), and Director General, Indian Council of Agricultural Research (ICAR), New Delhi, was felicitated by ICAR scientists of Hyderabad, at the Academy, on conferment of Padma Bhushan award for his outstanding contribution in the field of Plant breeding, on February, 1998. On this occasion, Dr Paroda said that to propel the Indian agriculture into the 21st century, the quality, technical skills, and management of agricultural manpower must improve in consonance with the rapidly changing global needs. Human Resource Development is a necessary concomitant of all dynamic systems for stability and for attaining equilibrium with external forces. Despite three fold increase in population, the foodgrains production in India has increased four times. Obviously, institutional support coupled with appropriate human resources had been the major cradle of our green, white, yellow, and blue revolutions, he added. Dr Paroda also mentioned that to create a healthy work-culture, it is imperative on the part of any organization to ensure that its employees are duly recognized through an appropriate incentive-and-reward system, which could be in the form of adequate laboratory facilities, support for research schemes, and awards for outstanding work. Dr Paroda stressed the need for more linkages with private sector for progress in agriculture.



Earlier, Dr J.C. Katyal, Director, NAARM, while welcoming the gathering said that Dr Paroda is an able research administrator with determination and total commitment. He has been successful in bringing about positive changes in the National Agricultural Research System, de-

spite several constraints.

Dr Paroda was responsible for crop production, improvement and protection programmes in the Asia-Pacific Region, including seed development and crop research activities. Prior to the conferment of Padma Bhushan by Government of India, Dr Paroda received many prestigious awards. These include Rafi Ahmed Kidwai Memorial Prize in 1982-83 for cluster bean research. In the following year, the ICAR Team Research Award was presented to Dr. Paroda and his group for their outstanding contributions to forage sorghum improvement. In 1988, he received the award of the Federation of Indian Chambers of Commerce and Industry (FICCI) for outstanding work in agriculture. The Prime Minister of India presented him the prestigious Om Prakash Bhasin Award in 1992 for outstanding achievements in agriculture. He is also a recipient of AISMAN award.

Dr Paroda served on the Board of Trustees of the International Rice Research Institute (IRRI) during the period 1990-93. He was also on the Board of Management of several important organizations such as IARI, NSC, and SFCI. He also represented India on a number of official delegations abroad. He has travelled widely to different parts of the world and also served in several international agencies. Dr R.S. Paroda was also the Chairman of the Governing Body of ICRISAT.

As a token of affection, Dr Katyal presented a memento to Dr Paroda, on behalf of the scientists of the ICAR institutes at Hyderabad. The Governing Body Members of ICRISAT, Dr S. M. Barghouti, Director General, ICRISAT, Dr E. A. Siddiq, National Professor, DRR, and Dr I.V. Subba Rao, VC, ANGRAU, Hyderabad, were also present on the occasion. Dr S.N. Saha, Joint Director, NAARM, proposed a vote of thanks.

## Guest lectures

### *Food Security meaningless without access : Swaminathan*

Building a sustainable food security system will be of no avail unless every individual in the country has "physical, economic, social, and environmental access" to a balanced diet, said Dr M.S. Swaminathan, Chairman of M.S. Swaminathan Research Foundation. Access to safe drinking water, sanitation, environmental hygiene, primary health care, and education were as essential as balanced diet for leading a healthy and productive life, he said while address-



ing the scientists of the Agric. Research Service at the Academy on June 18, 1997.

Dr Swaminathan, who was delivering a special lecture on the food security system, said that the efforts of agricultural scientists coupled with the hard labour put in by the millions of farmers made it possible to increase the foodgrain production from 51 million tonnes in 1951 to about 198 million tonnes in 1996-97. The yield of wheat and potato increased nearly nine times since 1951 and maize by six times. "What we have not been able to ensure is the economic access. The majority of the population is not having adequate income to buy the food they need, which is primarily due to population explosion", he said.

He cautioned against the increasing damage being done to the ecological foundation of agriculture such as land, water, forests, biodiversity, and atmosphere; and the distinct possibility of adverse changes in climate and sea level. Swaminathan emphasized the need for introduction of a well-defined regulatory mechanism, human resource development, re-organization of support systems, and ways and means to rejuvenate the National Agricultural Extension order to be able to meet the present and future challenges. An ever-green agriculture depends on promoting sustainable agriculture and rural development, conservation of biological diversity, environmental sound management of biotechnology, and protection of coastal areas and the oceans.

## Hindi Week Celebrations

Hindi week celebrations were organized at the Academy from September 8 to 14, 1997. The valedictory function was held on September 23, 1997. To mark this occasion, Hindi competitions like essay writing, typing, translation of words, elocution, and singing, etc. were organized for the staff of the Academy. Prizes were distributed to the winners of these competitions in the concluding function. Cash awards were also given to the staff who secured meritorious marks in the Hindi examination held during May 1997.



Dr. B. B. Bhatnagar, Director, NAARM, chaired the function. Dr Noorjahan Begum, Professor, Osmania University, Hyderabad, was the chief guest of the function. Dr S.N. Saha, Joint Secretary and the Chairman of the Official Language Implementation Committee of the Government of India, was also present on the occasion. Dr A. Gopalam, Officer-in-charge, Hindi,

presented the report on the achievements of the Hindi Section of the Academy during the year 1996.

## Foreign Visits

- \* Dr R. Kalpana Sastry, Senior Scientist attended the Fourth International Safflower Conference organised jointly by Food and Agricultural Organisation and the Agronomy Institute of Bari University, Agricultural Biology Department, University of Basilicata, Potenza, Bari, from June 2 to 7, 1997. She presented a keynote paper on "Managing pests of safflower - New paradigm", in the session on pest management. Her travel was sponsored by FAO under the programme on use of experts for technical cooperation among the developing countries.
- \* Mrs P.V. Nirmala, Technical Officer, attended the course on "Current trends in agriculture, forestry and natural resources information", organised by CAB International at the University of Putra, Serdang, Malaysia from September 29 to October 10, 1997.
- \* The Academy is entrusted with a national mandate for improving managerial competence in the country's agricultural research and education system. To meet this requirement the academy's faculty is bounded to design and offer quality training programmes to agricultural scientific personnel in the country in the area of educational technology and research management. To meet this requirement, provision has been made in the Agricultural Human Resources Development (AHRD) Project launched by ICAR in 1995 with support from World Bank to train the Academy's faculty abroad in the frontier areas of agricultural research and education management. Dr D. Rama Rao, Senior Scientist and Mr B. Pandu Reddy, Finance and Accounts Officer were deputed to USA from January 12 to April 10, 1998 for training at Ohio State University. Dr Rama Rao attended the courses on Decision Support and Expert Systems, Artificial Intelligence- Basic techniques; Women, Technology and Education; Educational Telecommunications; Computer applications in education; Database system; and Social and Ethical issues in computing etc. Mr Pandu Reddy attended the courses on Educational Finance; Business administration of schools; Accounting and management information systems; and Auditing - Principles & Procedures, etc. Dr Ajit P. Maru, Senior Scientist, was also deputed to attend three months training on development of "Global learning centres, virtual universities and virtual information centres" at Michigan State University, East Lansing, USA from March 20, 1998 under the AHRD project.

## Golden Indefest

The 50 years of Indian Independence was celebrated in a grand manner at the Academy. Golden Jubilee Independence Day Festival (Golden Indefest) was marked by the flag hoisting by Dr J.C. Katyal, Director, NAARM. Competitions were held for the members

of NAARM family in various literary, cultural, and sports events. In addition to the various games and sports events conducted, essay writing, quiz, just a minute, poetry recitation for children, antakshari, rangoli, and fancy dress competitions deserve a special mention. The competitions were marked by the enthusiastic participation of a large number of NAARM family members. Some competitions were also organized on the Independence day. Prizes were distributed to the winners of various competitions on the Golden Indefest day.

## Pulse Polio Camp

NAARM Health Centre organized Pulse Polio Immunization Programme on December 7, 1997, as a part of the Govt. campaign of polio eradication from our country. As many as 51 children below five years age group, residing in and around NAARM campus, were immunized under this programme.

## Vanamahotsavam

Vanamahotsavam was organized at the Academy on July 12, 1997. Inaugurating the function Dr J.C. Katyal, Director, NAARM, described trees as the lungs of our planet, as they purify the air we breathe in. He said that it is the duty of the person who plants a sapling to tend it so as to recall finer moments of the past when the tree grows. He added that planting trees will beautify one's surroundings, and will reduce pollution.

Dr Virupakashappa, Director, Directorate of Oilseeds Research; Mr P. Venkateswarlu, Forest Manager, and Mr Azizullakhan, Deputy Range Officer of Urban Forestry; staff members of Central Bank of India, NAARM Extension Counter; and members of the ICAR sister institutes residing in NAARM campus participated in the programme, in addition to the members of the entire NAARM family. Two thousand seedlings were planted on that day at the NAARM campus.

## HUMAN AND FINANCIAL RESOURCES

### Human resource

A summary account of human resources sanctioned for NAARM is presented in the following table. Information on vacant positions is also given.

| Category       | Sanctioned Strength | Posts Filled (as on 31-3-98) | Vacant Positions |
|----------------|---------------------|------------------------------|------------------|
| Scientific     | 43*                 | 21                           | 22               |
| Technical      | 66                  | 64                           | 2                |
| Administrative | 50                  | 47                           | 3                |
| Supporting     | 41                  | 41                           | -                |

\*Includes three RMP positions

During 1997, ICAR assigned specific subjects to sanctioned scientific positions at NAARM. Subject and post-wise approved categorization is as follows:

| Discipline                  | Cadre Strength |                  |                     |       |
|-----------------------------|----------------|------------------|---------------------|-------|
|                             | Scientist      | Senior Scientist | Principal Scientist | Total |
| Agricultural Research Mgmt* | -              | 4                | 4                   | 8     |
| Agricultural Statistics     | -              | 1                | 1                   | 2     |
| Computer Applications       | -              | 2                | 2                   | 4     |
| Educational Psychology*     | -              | 1                | 1                   | 2     |
| Educational Technology*     | -              | 3                | 2                   | 5     |
| Mass Communication*         | -              | 1                | -                   | 1     |
| Agricultural Extension      | -              | 3                | 1                   | 4     |
| Rural Sociology*            | -              | -                | 1                   | 1     |
| Human Resources Management* | -              | 4                | 1                   | 5     |
| Agricultural Economics      | -              | 2                | 1                   | 3     |
| Research Project Management | -              | 1                | 4                   | 5     |
| Total                       | -              | 22               | 18                  | 40    |

\*Discipline does not comply with the disciplines of Agricultural Research Service

## Financial resource

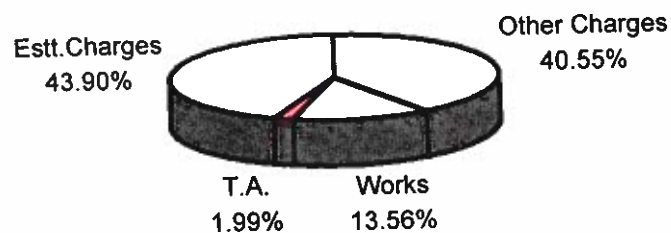
The budget allocation and expenditure during 1997-98 are given in the following table. Information presented in table below also provides details on finances available from other sources and revenue generated during the year under report. (Rs. in lakhs)

| Scheme                | Budget allocation | Expenditure |
|-----------------------|-------------------|-------------|
| <b>NON-PLAN</b>       |                   |             |
| Establishment Charges | 64.68             | 164.68      |
| Travelling Allowances | 2.50              | 2.50        |
| Other Charges         | 59.80             | 59.80       |
| Works                 | --                | --          |
| Sub-Total             | 226.98            | 226.98      |
| <b>PLAN</b>           |                   |             |
| Establishment Charges | 0.31              | 0.31        |
| Travelling Allowances | 4.95              | 4.95        |
| Other Charges         | 92.20             | 92.20       |
| Works                 | 50.84             | 50.84       |
| Sub-Total             | 148.30            | 148.30      |
| <b>Other Sources</b>  |                   |             |
| Summer Institute      | 1.75              | 0.88        |
| A.P.Cess              | 4.74              | 4.74        |
| AHRD                  | 153.44            | 86.99       |
| NATP                  | 3.00              | 2.32        |
| PWA                   | 2.50              | 3.08        |
| Sub-Total             | 165.43            | 95.67       |
| Grand Total           | 540.71            | 470.95      |

### Resource Generation (Rs. in lakhs)

Revenue generated at the Academy 27.62

### Proportional (%) Expenditure, 1997 - 98 (Plan + Nonplan)



## COMMITTEES

### Management Committee

The Academy is guided by the Management Committee in planning, organizing and conducting its various programmes. The Management Committee approves various programmes and financial outlays thereupon. The Academy ensures their implementation, both in terms of physical targets and schedules within sanctioned budget heads. The Management Committee also supports and guides in formulation and consideration of proposals for Five Year Plan and Annual Plans and takes up periodical review of progress of development schemes. During the year under report, two meetings of the Management Committee (i.e., on 26-04-97 and 07-03-98) were held.

The composition of the Management Committee (up to 17-9-97) of NAARM is as follows :

#### Chairman:

1. **Dr J.C. Katyal**, Director, NAARM

#### Members:

2. **Mr Anand Rau**, Principal Secretary, Dept of Agri & Co-op, Govt of A.P. Hyderabad, A.P.
3. **Dr Harpal Singh**, Dean, P.G. Studies, G.B. Pant Univ of A & T, Pantnagar, U.P.
4. **Dr M.V. Rao**, Vice-Chancellor, Acharya N.G. Ranga Agricultural University, Hyderabad, A.P.
5. **Ms K. Bharathi Devi**, Social worker, Kakinada, A.P.
6. **Mr Mohd. Haneef Khatri**, General Secretary, District Congress Committee, Sikar-Rajasthan.
7. **Dr O.P. Pareek**, Director, NRC for Arid Zone Horticulture, Bikaner, Rajasthan.
8. **Dr Dayanatha Jha**, Director, NCAP, New Delhi.
9. **Dr Mangala Rai**, Deputy Director General (Crops), ICAR, New Delhi.
10. **Dr K. Krishnaiah**, Director, DRR, Hyderabad, A.P.
11. **Dr S.L. Mehta**, Deputy Director General (Education), ICAR, New Delhi.
12. **Mr P.V. Eapachan**, F and AO, CRIDA, Hyderabad, A.P.

#### Member-Secretary

13. **Mr M. Suresh Kumar**, CAO, NAARM, Hyderabad, A.P.

## Management Committee

With effect from **18-9-97** composition of NAARM Management Committee is as follows:

### Chairman:

1. **Dr J.C.Katyal**, Director, NAARM, Hyderabad, A.P.

### Members:

2. **Mr Jannath Hussain**, Secretary, Dept of Agriculture, Govt of A.P., Hyderabad, A.P.
3. **Dr S.Kannaiyan**, Dean, Postgraduate Studies, TNAU, Coimbatore, T.N.
4. **Dr I.V.Subba Rao**, Vice-Chancellor, Acharya N.G.Ranga Agricultural University, Hyderabad, A.P.
5. **Prof B.D.Joshi**, R.K.College, Madhubani, Bihar.
6. **Ms Sabita Das**, Ex-MP, Meharpur (P.O), Silchar District, Kachar, Assam.
7. **Dr O.P.Pareek**, Director, NRC for Arid Horticulture, Bikaner, Rajasthan (up to 29-5-98).
8. **Dr Dayanatha Jha**, Director, NCAP, New Delhi – 110 012 (up to 29-8-98).
9. **Dr Mangala Rai**, Deputy Director General (Crops), ICAR, New Delhi (up to 29-5-98).
10. **Dr K.Krishnaiah**, Director, DRR, Hyderabad, A.P.
11. **Dr S.L.Mehta**, Deputy Director General (Education), ICAR, New Delhi.
12. **Mr B.Pandu Reddy**, F and AO, NAARM, Rajendranagar, Hyderabad, A.P.

### Member-Secretary:

13. **CAO**, NAARM, Hyderabad, A.P.

## Research Advisory Committee

The Research Advisory Committee (RAC) was constituted by the ICAR with the sole purpose of supporting and guiding NAARM in design and development of appropriate research programmes in the area of Agricultural Research and Education Management. During the year under report, an RAC meeting was held on September 19, 1997. Besides reviewing the overall research activities of the Academy, research projects approved by the SRC in May 1997 were discussed and finalised. The following is the current constitution of NAARM RAC:

**Chairman:**

1. **Dr Maharaj Singh**, Ex-DDG (Education), Gurgaon, Haryana.

**Members:**

2. **Dr S.L. Mehta**, Deputy Director General (Education), ICAR, New Delhi.
3. **Dr Y.L. Nene**, Ex-DDG, ICRISAT & Chairman, Asian Agri-History Foundation, Hyderabad, A.P.
4. **Dr Anil K. Gupta**, Professor (CMA), IIM, Ahmedabad, Gujarat.
5. **Dr K.S. Ramesh**, Dean of Studies, ASCI, Hyderabad, A.P.
6. **Dr Dharani P. Sinha**, President & MD, COSMODE, Hyderabad, A.P.
7. **Dr J.C. Katyal**, Director, NAARM.

**Member-Secretary:**

8. **Dr T. Balaguru**, Principal Scientist, NAARM.

## **Staff Research Council**

The Staff Research Council (SRC), deliberates on the research activities in the areas of agricultural research and education management. SRC comprises of all faculty members, some ICAR nominated members with Director, NAARM as its chairman. SRC meeting was organized on May 23 and 24, 1997 to review and approve the new project proposals prepared by the faculty members in the specialized areas of agricultural research and education management.

## **Official Language Implementation Committee**

The Official Language Implementation Committee of the Academy provides periodical suggestions for implementation of Official Language Policy, and rules and regulations of the Government of India that are issued from time to time. The Committee met two times during the year on June 26 and September 30, 1997, and discussed several issues relating to the implementation of Hindi language. The following is the constitution of the Official Language Implementation Committee:

**Patron:**

1. **Dr J.C. Katyal**, Director

## Feedback on training programmes

As outlined in the Chapter on "Training programmes" besides FOCARS, NAARM organizes some 40 programmes every year. Of these, some programmes are of recurrent nature, while others are conducted in accordance with the demand and client needs. Irrespective of the nature, gathering feedback through structured questionnaire from the participants, on conduct, content, relevance and applicability is a regular feature of each programme. Comments/suggestions, thus received, are collated and synthesized into recommendations for revising prevalent methodologies/approaches on organization and delivery of various courses to introduce future improvements. A brief account of the feedback received on various training programmes, conducted by NAARM during 1997-98, is presented below:

| Programme                      | Feedback                                                                            |                                                                                                                                                                                                            |
|--------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | Evaluative                                                                          | Corrective                                                                                                                                                                                                 |
| <b>FOCARS</b>                  | Project development and project proposal preparation                                | Less time for coverage of administrative and financial rules                                                                                                                                               |
|                                | Practical training in the use of audio-visual aids                                  | Guest lecture proportion inadequate                                                                                                                                                                        |
|                                | Hands on experience with computers                                                  | Shorter duration and its consequent telling on tight schedule.                                                                                                                                             |
|                                | Punctuality and regularity of conduct.                                              | FET needs reorientation on its more effective organization.                                                                                                                                                |
| <b>Senior-level programmes</b> |                                                                                     |                                                                                                                                                                                                            |
|                                | New learning experience to manage agricultural research more effectively            | Need to sensitize Directors and other RMPs on the concepts and usefulness of training. This would create a supporting climate for the participants to put to use the knowledge gained from various courses |
|                                | Good reading material provided                                                      |                                                                                                                                                                                                            |
|                                | Computer-based capsules built confidence in handling various packages independently |                                                                                                                                                                                                            |

Presentation substance and style, and conduct of faculty

Course may be made broad based to include corporate sector and NGOs

Introduce more case studies to build problem-solving confidence

### Faculty development programmes

Course curricula is an eye opener to the science and art of teaching

Use of advanced instructional aids is inadequate and not up to the mark

Commitment, confidence and competency of the faculty members is exemplary

More emphasis necessary on manpower planning, financial management, resource generation, experiential learning, teaching effectiveness and personality development

### Sponsored programmes

Applicability and adaptability of the programme to suit the requirements

Supply of relevant books and software far below the minimum needs

Adequate provision on exercises, hands-on experience and group learning

Non-supply of lecture notes of guest speakers

### Workshops

Methodical organization

Examples involving Indian case studies remain inadequate

Offer valuable forum to interface and appreciate scientist-managers ideas and viewpoints

Shorter duration; can be longer for programmes of topical interest

Topic selections relevant to present day needs

## International programmes

Topic selection relevant to the programme and participatory delivery approach is novel

Study visits are highly beneficial

Team spirit of NAARM staff – an example to emulate

Inadequate time allotment for discussion and hands-on experience

Duration of programme requires extension

Non-availability of continental kind of food

# RESEARCH

## Research projects

The Academy is mandated to conduct research in the areas of agricultural research and education management. Its scientists carry out research on various aspects of agricultural policy, research project management, transfer of technology, human resources development, agricultural education, and information technology. During the year under report, on approval of RAC, a decision was taken that each faculty member will devote about 30 per cent of time on research. A preliminary description of various research projects, initiated during 1997, is presented below :

- **Management and Impact of Agricultural Research Stations/ Institutes in India**  
(K. Venkateswarlu)

The project aims at studying the processes and details of technology developed in agricultural research stations and also the productivity of various research / experimental stations. The constraints faced in the management of research stations would also be found out. The schedules have been developed. Case studies have been initiated and data are being collected for this purpose. The work is under progress.

- **Criteria for Selection and Formulation of R & D Projects in Agriculture and Allied Areas**  
(K. Venkateswarlu and S.K. Nanda)

The project aims to study the criteria for selection of R & D projects in agriculture and in allied areas, and also investigate the formulation process of various R & D projects. Schedules / questionnaires have been developed and data are being collected.

- **Creativity and Problem Solving Nature of Agricultural Scientists**  
(K. Venkateswarlu and N. Sandhya Shenoy)

The main objective of this study is to understand the creativity and problem solving nature of agricultural scientists in developing new theories / inventions / discoveries in the field of agricultural sciences. The data have been collected from the scientists and senior scientists through a standard questionnaire. The work is under progress.

- **Management and Impact of All India Coordinated Research Projects**  
(K. Venkateswarlu and S. Shanmugam)

The main objective of the project is to identify the lacunae in the structure and organization of the All India Coordinated Research Projects (AICRPs) and also to study the impact of various AICRPs. A questionnaire has been designed and mailed to project coordinators to collect the data on these aspects. Data have been received from ten AICRPs and are being processed.

● **Impact of National Agricultural Research Project**  
(K. Venkateswarlu, K. Vidyasagar Rao and B.S. Chandel)

The project aimed to develop a statistical model for measuring the impact of NARP in 120 agro-climatic zones. The data have been collected from RARS, Lam from ANGRAU. Some more data collection is in progress.

● **Research Project Management in the National Agricultural Research System**  
(P. Manikandan, T. Balaguru and S.N. Saha)

The project aims to study the existing system of project planning and implementation in the NARS, identification of project areas and objectives; preparation of research proposal; review of research proposals; approval of research projects and commitment of resources; implementation of research and to identify the time and resource budgeting practices followed and to assess their effectiveness. The literature survey on the subject has been initiated.

● **Professional (Graduates) Women in Agriculture (PWA)**  
(D. Rama Rao, N. Sandhya Shenoy and Mrs Suguna Sri)

A software package was developed to maintain database on graduate women in agriculture and also organizations employing them. Programme for data entry has been finalised. The responses have been codified for easy computation and analysis. The codified data entry is in progress.

Based on the problems identified during the brainstorming session and also from the introspectual responses received from the subjectors involved in earlier studies, a questionnaire was designed to capture information from women graduates on issues such as education, family, social and economic backgrounds, employment, professional achievements and on some specific issues pertaining to the problems faced by women.

● **Impact of NAARM training on ICAR Scientists and assessment of their training needs (1988 - 1994)**  
(K. Vidyasagar Rao and N. Sadhya Shenoy)

The project aims to evaluate the impact of NAARM training on ICAR scientists and their environment, to find out the extent to which the NAARM training had helped in job performance of scientists in their work environment and finally to document the strengths and weaknesses of the NAARM training programmes and evaluate the training needs of the ICAR scientists. the work is in progress, only the proforma for collecting the data has been finalised.

- **Role of NGO and GO KVKs in Agricultural Development with special reference to Andhra Pradesh**  
(N. Sandhya Shenoy and K. Vidyasagar Rao)

The major objectives of the study are to study and compare the selected organizational, workgroup, and job involvement variables and roles performed in agricultural development by the GO and NGO KVKs and also to find the inter-institutional linkage patterns among the NGO KVKs and other line departments, their expectations, operational constraints and suggestions for meaningful collaboration. During the period under report, the literature collection and data gathering were initiated.

- **A profile of Scientific Human Resources in ICAR System in India**  
(S. Shanmugam)

The project was undertaken to study the socio-economic attributes of cadre strength of scientific personnel in ICAR Institutes, and to plan the optimum human resources profile in ICAR system for 2001 A.D. The data collection is in progress.

- **Evolving a Suitable Methodological Framework for Priority Setting in Agriculture**  
(T. Balaguru)

The project was initiated with the objective of developing suitable methodology for setting research priorities in agriculture. The data collection is in progress.

- **Implementation of Farming Systems Research (FSR) in India : Case Studies**  
(T. Balaguru and P. Manikandan)

The project was undertaken to identify the leadership styles of project leaders in ICAR and measure their effectiveness and also to evolve measures to improve the performance of research teams through effective leadership development. At the time this report is being finalised, the proforma for basic data collection was under preparation.

- **Technological Forecasting of Future Oilseeds Scenario in India**  
(D. Rama Rao, R. Kalpana Sastry and V. Kiresur)

The two-year duration AP CESS fund ad-hoc research project was initiated on April 1, 1997 in collaboration with Directorate of Oilseeds Research. The study aims to identify constraints and potentials that will influence the country's oilseed scenario in the future by employing systems approach coupled with technological forecasting.

During the year under report, the infrastructure facilities like purchase of computer equipment and recruitment of research associates that are required were completed. Under technical programme, a thorough review of global and Indian oilseeds scenario was made

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and based on that an overview report was also prepared which includes the past events, present status and future strategies of almost all the areas associated with oilseeds. Creation of a database on institutions/agencies/individuals involved in each of the areas associated with oilseeds is in progress. A questionnaire, which is under pre-testing, was prepared keeping in view the objectives of the project and also the issues raised in brainstorming session held on Dec. 12, 1997 at the Academy with about 25 experts from different organizations associated with oilseeds. The major sub-issues raised during the brainstorming session were:

- Government policies with regard to R&D, finance/subsidies, processing, prices and import and export strategies.
- Under R&D mainly the knowledge gaps and need for exploration of the same.
- Competing crops.
- Marketing problems.
- Major constraints of processing / private sector.
- Changing consumer preferences, etc.

Forecasting and assessment of future potential, development of alternative scenarios based on historic data and forecasts, systems evaluation will be undertaken.

## CONSULTANCY

The consultancy is supposedly a major activity to further academic pursuits and to simultaneously generate resources in today's highly competitive world. It also enables to optimize the full potential of the expertise by undertaking several sponsored, contract research projects for the national and international institutions and agencies, and non-governmental organizations. The faculty members of the Academy undertook consultancy on cropping systems development under command area programme and organization of off-campus programmes in research and education management for the SAUs. During the year under report, a few consultancy projects were undertaken at the request of certain national and international agencies. Under the auspices of the AHRD project, supported by the World Bank, consultancy work was undertaken in areas such as priority setting in agricultural research, and education technology. The following consultancy services were provided:

### Off-campus Programme in Sikkim

A Trainers' Training Programme was organized by the Academy in Sikkim by Drs T. Balaguru and P. Manikandan from June 23 to July 5, 1997 with the support provided by the Indo-Swiss Project - Sikkim (ISPS). The participants of this course belonged to the Horticultural Department of Sikkim. The main objective of the course was to build capacity of the participants while transferring technologies on ginger disease management. The participants were exposed to various aspects of training and were helped in the design of a training package on ginger disease. Using the developed package, the participants were encouraged to practice through a two-tier field training.

### Consultancy to Department of Animal Husbandry

The Academy provided consultancy in planning the Livestock Development Information System (LDIS) to the Department of Animal Husbandry, Andhra Pradesh (DAH-AP). The project on developing the LDIS was sponsored by the Indo-Swiss Project - Andhra Pradesh (ISPA). The main objective of the project was to develop concepts and tools for the introduction and for further strengthening of planning and management of livestock development programmes and related information systems; to strengthen MIS application by AHD staff for planning, implementation, maintenance, and monitoring of programmes; and also for decision making.

The Livestock Development Information System is a decision support system for planning and monitoring of livestock development programmes and services at various levels according to the organizational structure of the State Department of Animal Husbandry. The LDIS is planned to correct problems related to timely availability, access, and reliability of data and information related to animal husbandry activities. It comprises five modules to maintain

household and village level livestock population data and information on veterinary infrastructure, animal improvement programmes, disease surveillance and monitoring, and forecasting fodder availability. This is being done by improving data collection at the grassroots and also by developing analytical and usage skills for "informed" decision making. The LDIS will improve the impact of livestock development programmes, providing livestock services and disaster management.

The development of the LDIS is being facilitated, alongwith staff of DAH and ISPA, by two scientists from the Academy, Drs Ajit Maru and D. Rama Rao.

## Consultancy to ODA

Dr J.C. Katyal, Director, NAARM provided consultancy to Overseas Development Administration (ODA) (now Department for International Development, DFID), United Kingdom. Purpose of the consultancy was to review the preliminary design of the follow on project of the ongoing ODA supported KRIBHCO Rainted Farming Project (KRIBP/West) which is based in the three contiguous districts of Panchmahals (Gujarat), Banswara (Rajasthan) and Jhabua (MP). Terms of Reference for the Design Review Team were:

- \* to review, at a general level, the overall design of the project, assess whether the approaches proposed are likely to be effective and recommend any changes;
- \* to review the proposed linkages with GOs, NGOs, PRIs, ICAR institutes and SAUs in different states and the ways the project will work with them (these may vary among states), identify possible risks, and recommend changes, if necessary;
- \* to develop criteria for selection and funding of partner GOs, NGOs, PRI, ICAR institutes and SAUs and identify possible target organizations;
- \* to recommend how, in a step-by-step way, the project should build and maintain linkages with partner GOs, NGOs, PRIs, ICAR institutes and SAUs;
- \* to review the draft logical framework (to be provided by the Design Team) and identify possible objectively verifiable indicators (OVIs) for the purpose and the output(s) dealing with linkages with other organizations;
- \* to recommend, in light of the new management systems proposed for the KRIBHCO projects, the nature and involvement of GOs, NGOs, PRIs, ICAR institutes and SAUs in project management and governance;
- \* to recommend appropriate strategies to integrate project villages with the process of area planning through PRIs envisaged under the 73<sup>rd</sup> Constitutional Amendment Act, identify any risks and suggest ways to minimize them;

- \* to recommend measures to ensure the sustainability of the new approaches by GPs, NGOs, PRIs, ICAR institutes and SAUs in the three states;

Consultancy Report was submitted to Team Leader which was accepted by ODA (now DfID) – Department for International Development).

### **Consultancy to Rajasthan Agricultural University**

At the request of Academic Staff College of Rajasthan Agricultural University, Dr A. Gopalam has provided the consultancy support for the development of their staff by conducting a capsule training programme in curriculum design and development, teaching effectiveness and lesson planning.

## Awards

### Faculty

Dr D. Rama Rao, Sr. scientist and Mr U. Muralidhar, programmer (computers). NAARM received the Hari Om Ashram Trust award for their outstanding research contributions in database management of agricultural research and education, for the biennium 1995-96, from the Hon'ble Union Agricultural Minister Sri Chaturanan Mishra in New Delhi on July 16, 1997. They have been successful in generating a reliable database on the SAUs information system (AGRIUNIS) under the project commissioned by the Department of Science and Technology (DST).

Drs T. Balaguru, Jagannadham Challa and P. Manikandan, Principal Scientists of the Academy were adjudged as excellent teachers by the Education Division of ICAR, New Delhi, based on the feedback response from the scientist-probationers of FOCARS programmes. It is for the first time such a recognition was bestowed on the members of the NAARM faculty.

### Best institutional rose garden

The Academy bagged the title of best Institutional Rose Garden, King of the Show, Prince of the Show, and Best Red Rose, besides the second best display of roses by the institution at XI Annual Rose Show, organized by the Secunderabad Horticultural Society at the Contonment Garden on December 20 and 21, 1997. The Academy also won 13 prizes, which include six first prizes and seven second prizes under different sections of flower display.

In the XXII Annual Rose Show conducted by the Hyderabad Rose Society at the Public Gardens on 13th and 14th December, 1997, the Academy bagged the prestigious trophies for Prince of the Show and for the best display of roses. Apart from the above, it bagged 26 different prizes (Six first prizes, twelve second prizes, and nine third prizes) under different sections of flower display.

### NAARM's share of prizes in the South Zone Sports Meet

The NAARM sports contingent, consisting of 16 members including two women, led by Dr A. Debnath, Manager, took part in the ICAR inter-institutional Zone III Sports meet organized by Sugarcane Breeding Institute, Coimbatore, from November 5 - 10, 1997. NAARM topped in the medal tally won among the ICAR institutes located in Hyderabad. Mrs Rukmani Ammal got 1st prize in shot put and third prize in discus throw. Mrs G. Aneeja got second prize in javelin throw and, Dr A Debnath got first prize in men's javelin throw.

## Agricultural Human Resources Development Project

Agricultural Human Resources Development project (AHRD) consists of development of educational technology cells, strengthening of existing facilities at NAARM, and conducting teacher improvement programmes at various Agricultural Universities.

For the teacher improvement programme component, seven training programmes were conducted for the faculty of TNAU with the project assistance from the university at three levels. One at assistant and associate professors level, second at professors and senior professors and third at the level of directors, deans, heads of the research stations and departments. The feedback indicated that such training programmes were useful for improving the teacher qualities both at instruction and behavior levels and also for improving the quality of agricultural education. The objectives set for these training programmes were:

- \* to train the teachers in agricultural education management
- \* to strengthen educational initiatives of teachers for optimizing learning teaching in TNAU
- \* to bring in behavioral modifications by organising training in attitude related issues.

A senior level workshop on "How to manage agricultural education better?" was also conducted for the Deans of the State Agricultural Universities.

For strengthening the existing facilities at NAARM both video and computer equipment were procured at different stages. The establishment of educational technology cells is confined to organization of workshops and training programmes at different levels of educational planners, administrators and functionaries.

## Linkages

Besides establishing linkages with several national institutes, NAARM made concerted efforts to build working relationships with relevant CGIAR institutions in the specialized area of agricultural research management. With the mandate similar to that of NAARM, the International Service for National Agricultural Research (ISNAR) mooted the suggestion on establishing an ICAR-ISNAR Regional Centre at NAARM, as a mutually rewarding exercise. Incidentally, setting up of ICAR-ISNAR Regional Centre is not altogether new, it has its conceptual roots in an old idea on creation of mini-ISNARs in each region of the developing world. It was only during the March of 1998, when Directors General of ICAR and ISNAR met at the TAC meeting in ICRISAT and agreed to revive this novel concept of working together. Coming to the close heels of this meeting, collaboration between ISNAR and ICAR, in general, and, NAARM in particular, was explored further during the visit of Director General

**Chairman:**

2. Dr S.N. Saha, Joint Director

**Vice-Chairman:**

3. Dr A. Gopalam, Principal Scientist

**Members:**

4. Dr D. Rama Rao, Senior Scientist
5. Mr M. Suresh Kumar, Chief Admn. Officer
6. Mr B. Pandu Reddy, Finance & Accounts Officer
7. Dr Jagannadham Challa, Principal Scientist
8. Dr S. Shanmugam, Senior Scientist
9. Dr M.M. Anwer, Senior Scientist
10. Dr B.S. Chandel, Senior Scientist
11. Ms T. Vanisri, Stenographer
12. Ms D. Dhana Laxmi, Lib. Asst.

**Member-Secretary:**

13. Mr D. Venkateswarlu, Tech. Officer (Hindi).

## VISITORS

### Hon'ble Union Minister of State for Agriculture, Dr S. Venugopalachary visits the Academy



The management of agricultural manpower should improve in consonance with the changing needs of society to take Indian agriculture into the 21st century, said the Hon'ble Union Minister of State for Agriculture, Dr S. Venugopalachary, while addressing the scientists on September 14, 1997, at NAARM. He exhorted the scientists to be ready for facing the challenges on the food front. He expressed concern at the population explosion, which was nullifying the remarkable progress achieved in agriculture.

### Other visitors

- ❑ A batch of 58 students of final year B.Sc. (Ag) of Pandit Jawaharlal Nehru College of Agriculture, Karaikal, visited the Academy on April 2, 1997.
- ❑ A group consisting of 16 Sri Lanka Extension and Research Officers visited the Academy on April 4, 1997.
- ❑ A batch of 69 students of final year B.Sc. (Ag) of Tamil Nadu Agricultural University, Coimbatore, visited the Academy on April 22, 1997.
- ❑ Dr Peter Smith from U.K., and FAO Consultant for ICAR, visited the Academy from May 19 - 21, 1997, to discuss projects under NATP. Scientists from ICAR institutes and SAUs, undertaking NATP - supported projects participated in the meetings.
- ❑ A three-member delegation from Yemen, consisting of Dr A. Dameen, Director General, Research; Mr Mohiyuddin El-Ghori, Director, Field Crops; and Dr S.V.R. Shetty, Team Leader; ICARDA, Yemen, visited the Academy on July 8, 1997. The delegation had discussion with the faculty members of the Academy. It also looked at the possibility of organizing training programmes for them in the areas of management. The team evinced keen interest in NAARM and expressed an intention to utilize the training and consultancy services for their country.

- ❑ Dr S. Gouro Abdoulaye, Director General, Institute of National d Aecherches Agronomiques du Niger (INRAN), Niamey, Niger, visited the Academy on August 12, 1997.
- ❑ A group of four IAS probationers of the 1996 batch, who have been allotted to Andhra Pradesh and undergoing training with Dr MCR Institute of Administration, visited the Academy on August 13, 1997.
- ❑ A group of 69 students of final year B.V.Sc. of Tamil Nadu Veterinary and Animal Sciences University, Chennai, visited the Academy on September 1, 1997.
- ❑ A group of 60 students of final year B.Sc. (Ag) of Annamalai University, Annamalai Nagar, Tamil Nadu, visited the Academy on September 15, 1997.
- ❑ A group of 60 students of final year B.V.Sc. of Tamil Nadu Veterinary and Animal Sciences University, Namakkal, visited the Academy on September 17, 1997.
- ❑ Two batches of 70 students each of B.V.Sc. (final year) of Annamalai University, Annamalai Nagar, visited the Academy on September 21 and 22, 1997.
- ❑ A group of seven editors, and senior correspondents of various newspapers and magazines from Meghalaya visited the Academy on October 21, 1997.
- ❑ A group of twenty eight final year B.Sc. (Home Science) students of Rajasthan Agricultural University visited the Academy on October 23, 1997.
- ❑ A press party consisting of seven editors and senior correspondents, belonging to Bangalore, visited the Academy on November 4, 1997.
- ❑ Dr H.M. Eisa, Senior Agriculturist, The World Bank, IBRD, International Development Association, Resident Mission in India, New Delhi, visited the Academy on November 18, 1997.
- ❑ Dr Alexander Vonder Osten, Executive Secretary, Consultative Group on International Agricultural Research (CGIAR), Washington DC, USA, visited the Academy on November 18, 1997.
- ❑ Dr S.M. Barghouti, Director General, ICRISAT and Dr Winkerman, Chairman, TAC, ICRISAT, India, visited the Academy on November 26, 1997.
- ❑ A group consisting of six District Agricultural Officers from Kenya Agricultural Research Institute, Kenya, visited the academy on December 10, 1997.

- A group of 75 final year B.Sc. students of University of Agricultural Sciences, Dharwad, visited the Academy on January 12, 1998.
- A group of 10 Final year B.Sc. students of Forestry, from Mettupalayam, TNAU, Coimbatore, visited the Academy on January 20, 1998.
- A group of 16 Final year B.Sc. (Home Science) students of Tamil Nadu Agricultural University, Madurai, visited the Academy on January 24, 1998.
- A group of 20 final year B.Sc. (Ag) students of Kerala Agricultural University, Nileshtar, visited the Academy on February 11, 1998.
- Dr R.P. Sapkota, Director, Crops and Horticulture, Nepal Agricultural Research Institute, Nepal; Dr Z. Karin, Chairman, BARC, Bangladesh, and Dr Dhruva Joshy, Extension Director, NARC., Nepal, visited the Academy on February 13, 1998.
- Dr Suyanto, Director of Research Institute for Legumes and Tubers, Malang, Indonesia, and Mr Narong Sak Senanarong, Director of the Field Crops Research Institute, Bangkok, Thailand, visited the Academy on February 18, 1998.
- A group of 47 final year B.Sc. (Ag) students of Tamil Nadu Agricultural University, Coimbatore, visited the Academy in two groups on February 12 and February 27, 1998, respectively.
- A group of 50 final year B.Sc. (Ag) students of Pandit Jawaharlal Nehru College of Agriculture and Research Institute, Karaikal, visited the Academy on March 4, 1998.
- A group of 94 final year B.Sc. (Ag) students of Agricultural College and Research Institute, Killikulam, visited the Academy on March 4, 1998.
- A group of 54 final year B.Sc. (Ag) students of Agricultural College and Research Institute, Tiruchirappalli, visited the Academy on March 12, 1998.
- A group of 110 final year B.Sc. (Ag) students of Agricultural College and Research Institute, Madurai, visited the Academy on March 12, 1998.
- Dr Stein Bie, Director General, ISNAR, visited the Academy on March 26, 1998. Dr Bie discussed the possible collaboration with NAARM in the area of Distance Learning and creation of ICAR-ISNAR Regional Centre at NAARM.
- Brig B.Sc. Pawar, Commandant, ARTY Centre, Hyderabad, visited the Academy on March 30, 1998.

## Research / Review / Conference papers

Balaguru, T. 1997. Policy, organization and management issues in setting up an IARI. Paper presented in the ICAR - ISNAR International Workshop on Information Policy for Agril Research held at NAARM, Hyderabad during December 8 - 11, 1997.

Chandel, B.S., and Chakor, I.S. 1998. Economic evaluation of crop management research for farmer's adoption - A study on ginger cultivation in Himachal Pradesh. Indian Journal of Agricultural Economics. 53, No. 1 pp:53-60.

Gopalam, A., and Kalla, J.C. 1997. Conceptual frame work on planning and management of education in agricultural sector. In Learning Management. Vinaysheel Gautam (ed) pp 160-171.

Gopalam, A. 1997. Profile of essential chemical components in tree spices. In Cultivation and Utilization of Medicinal and Aromatic Plants. S.S. Handa and B.M. Kaul (eds). Published by the Regional Research Laboratories, Jammu Tawi. pp 405-424

Gopalam, A. 1997. By-product utilization in turmeric and ginger. In Cultivation and Utilization of Medicinal and Aromatic Plants. S.S. Handa and B.M. Kaul (eds). Published by the Regional Research Laboratories, Jammu Tawi. pp 383 - 404.

Katyal, J.C. ; Challa, J. (ed s) 1997. Decentralization and devolution of powers in ICAR : Proceedings of the Policy Level Workshop in a Retreat Mode. NAARM publication. Jan 1998.

Katyal, J.C., Sharma, K.L. and Srinivas, K. 1997. Sulphur in Indian agriculture, Paper published in TSI/FAI/IFA Symposium on Sulphur in Balanced Fertilization, New Delhi, Feb 13-14 pp KS-2/1 to KS-2/12.

Katyal, J.C., Sharma, K.L., Srinivas K. and M. Narayana Reddy. 1997 Balanced fertilizer use in semi-arid soils. Fertilizer News, 42(4): 59-69

Katyal, J.C. 1997 Integrated concept of sustainability. The Hindu Survey of Indian Agriculture pp 25-31.

Katyal, J.C and Balaguru, T. 1997 India's Natural resources vis-a-vis research priority setting. Paper presented at the International Conference on Agricultural Research Priority Setting, Monitoring and Evaluation in the Indian NARS at IARI, New Delhi, July 21 & 23.

Katyal, J.C., Karuna Chand, L. and Gangadhara Rao, D. 1997 Agroforestry in Andhra

Pradesh : Opportunities for Biotechnological interventions'. Paper presented at the Workshop on Biotechnology Interventions for the Improvement of Agroforestry, Horticulture and Fodder' held at IPE, Hyderabad, Nov 29 & 30.

Katyal, J.C., 1997. Indian agriculture - profile of land resources, crop performances and Prospects. Dr D.P.Motiramani Memorial Lecture, delivered at MPKV, Rahuri and published in Journal of Indian Society Soil Science. 45: 659-674.

Shivakumar, Omkar Singh, Van Rhunen, H.A, and Rao, K.V.S. 1997. Repeatability of different stability parameters in chickpea yield trials, Plant Breeding:116.

Shivakumar, Omkar Singh, Van Rhunen, H.A, and Rao, K.V.S. 1997. Correlation among different stability parameters for grain yields in chickpea. Indian J. Gen and Pl Breeding.57.

## Books

Kanwar, J.S., and Katyal, J.C. (eds) 1997. Plant nutrient needs, supply, efficiency and policy issues : 2000 - 2025. Proceedings of the NAAS symposium on Plant Nutrient needs, supply, efficiency and policy issues : 2000 - 2025 held at CRIDA, Hyderabad.

Katyal, J.C., and Farrington John (eds) 1997. Research for rainfed farming. Proceedings of the ICAR-ODA workshop on research for rainfed farming held at CRIDA, Hyderabad on Sept. 11-14, 1995.

## Resource papers developed

- \* ARFIS manual [version 2.2]
- \* Personality development of teachers - Jagannadham Challa
- \* Reading skills and comprehension - Jagannadham Challa
- \* Seven capsules were developed for the Faculty development programmes in Educational Technology under AHRD project by A. Gopalam and Jagannadham Challa. These are A. Learning; B. Teaching; C. Testing and evaluation; D. Curriculum; E. Management of agricultural education; F. Instructional aids and media effectiveness; G. Behavioural modification and learner teaching.
- \* A background paper on Decentralization and devolution of powers in ICAR. J.C.Katyal
- \* A background paper on Project Based Budgeting. J.C. Katyal, K.P.C. Rao, P. Manikandan, Mruthyunjaya and N.P. Rajashekar.
- \* Guidelines on Project Based Budgeting. J.C. Katyal, K.P.C. Rao and Mruthyunjaya.

J.C. Katyal, Director

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K. Venkateswarlu, Principal Scientist (ARM) & In-Charge  
 VE. Sabarathnam, Principal Scientist (AE)  
 T. Balaguru, Principal Scientist (ARM)  
 N. Sandhya Shenoy, Senior Scientist (AE)  
 B.S. Chandel, Senior Scientist (AEco)  
 S.K. Nanda, Senior Scientist (ARM)

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 Jagannadham Challa, Principal Scientist (ET)  
 A. Gopalam, Principal Scientist (EP)  
 S. Shanmugam, Senior Scientist (ARM)  
 M.M. Anwer, Senior Scientist (HRM)  
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 D. Rama Rao, Senior Scientist (ARM)  
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 Ajit P. Maru, Senior Scientist (CA)

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 Capt. Mehar Singh (w.e.f Mar.9, 1998)  
 P. Janakiramaiah, AAO

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P. Namdev, Technical Officer

### **Health Centre**

A. Debnath, Medical Officer

### **Farm Services**

R. Lalpana Sastry O I/c  
V. Murali, Farm Superintendent  
M.A. Basith, Technical Officer

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N.R. Nageswara Rao, Technical Officer

### **Hostel Services**

P. Kumar, O I/c  
Zameer Ahmed, Manager

### **Video Services**

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Ch. Janardhan Rao, Technical Officer

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D. Venkateswarlu, Technical Officer

### **Library Services**

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P.V. Nirmala, Technical Officer

## **Computer Section**

A.P. Maru, O I/c  
U. Muralidhar, Programmer

## **Publication Services**

D. Rama Rao, O I/c  
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N. Narasimha Reddy, Technical Officer



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